

Maf-family bZIP transcription factor NRL interacts with RNA-binding proteins and R-loops in retinal photoreceptors

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Ximena Corso-Díaz , Xulong Liang, Kiam Preston Jr., Bilguun Tegshee, Milton A English, Jacob Nellissery, Sharda Prasad Yadav, Claire Marchal, Anand Swaroop 

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States • Department of Ophthalmology, Byers Eye Institute, Stanford University, Stanford, United States • In silichrom Ltd, Newbury, United Kingdom

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eLife Assessment

This **important** study employed multiple orthogonal techniques and tissue samples to investigate the interaction between the NRL transcription factor and RNA-binding proteins in the retina. The findings are **convincing** to support an interaction between NRL and the DHX9 helicase. The significance of the study could be enhanced with functional experiments of NRL-R-loop interactions in the developing retina and their potential role in photoreceptor health and gene regulation.

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Abstract

RNA-binding proteins (RBPs) perform diverse functions including the regulation of chromatin dynamics and the coupling of transcription with RNA processing. However, our understanding of their actions in mammalian neurons remains limited. Using affinity purification, yeast-two-hybrid and proximity ligation assays, we identified interactions of multiple RBPs with NRL, a Maf-family bZIP transcription factor critical for retinal rod photoreceptor development and function. In addition to splicing, many NRL-interacting RBPs are associated with R-loops, which form during transcription and increase during photoreceptor maturation. Focusing on DHX9 RNA helicase, we demonstrate that its expression is modulated by NRL and that the NRL-DHX9 interaction is positively influenced by R-loops. ssDRIP-Seq analysis reveals both stranded and unstranded R-loops at distinct genomic elements, characterized by active and inactive epigenetic signatures and enriched at neuronal genes. NRL binds to both types of R-loops, suggesting an epigenetically independent function. Our findings suggest additional functions of NRL during transcription and highlight complex interactions among transcription factors, RBPs, and R-loops in regulating photoreceptor gene expression in the mammalian retina.

Introduction

Cell type-specific and quantitatively precise decoding of genetic information produces a plethora of divergent cellular morphologies and functions during development ^{1–3}. The initiation of DNA-dependent RNA synthesis by RNA Polymerase II is coordinated by complex interactions among chromatin modifying proteins, ubiquitous and cell-type specific transcription factors, and transcriptional machineries ^{4–6}. Combinatorial and sequence-specific binding of trans-acting regulatory proteins to *cis*-regulatory elements including enhancers and promoters is facilitated by specific genome topologies, resulting in productive transcriptional output by assembling phase-separated condensates ^{4,6}. Concurrent control mechanisms meticulously guide every step of the genetic program. Curiously, cell type- and tissue-specific transcription factors are also reported to contribute to additional transcriptional and post-transcriptional regulatory steps, including 3-D chromatin dynamics and splicing ^{7–9}, to achieve stringent spatiotemporal control of gene expression patterns.

Gene regulatory networks underlying neuronal differentiation in the mammalian retina ^{10–14}, especially photoreceptors ^{15–17}, offer an attractive system to dissect divergent roles of transcriptional factors in controlling quantitatively precise cell type-specific gene expression. The Maf-family basic motif leucine zipper (bZIP) transcription factor NRL determines rod photoreceptor cell fate as its loss leads to retina with only cone photoreceptors ¹⁸, whereas ectopic expression of NRL in developing cones results in generation of rods ¹⁹. Continued expression of NRL is essential for maintaining rod function ²⁰, and mutations affecting NRL activity are associated with retinopathies ^{21–24}. NRL establishes rod cell identity and function by regulating precise transcriptional output of thousands of genes ^{25–27} in collaboration with cone rod homeobox (CRX) ^{28–30} and other regulatory proteins ^{15,16,31,32}. Given its fundamental role as a primary activator of rod expressed genes and continuous high-level expression during the life of rod photoreceptors, we hypothesized the involvement of NRL in modulating additional steps during the transcription process. This prediction was strengthened by transcriptional activation synergy observed in rhodopsin regulation between NRL and NonO/p54^{nrb} ³³. NonO, a ubiquitous nuclear paraspeckle protein, binds to DNA, RNA as well as proteins, and regulates distinct cellular events including the coupling of the circadian clock to cell cycle and transcription to splicing ^{34–36}.

Here, we focused on identifying additional roles of NRL in guiding rod gene expression by characterizing NRL-interacting proteins from the mammalian retina using multiple complementary approaches. We discovered an over-representation of RNA-binding proteins (RBPs) among NRL interactors, with almost half of high confidence interacting proteins associated with R-loops, which are DNA structures comprised of RNA-DNA hybrids and displaced single-stranded DNA. We then investigated the interaction of NRL with two key RNA helicases, DHX9 and DDX5, which mediate resolution of R-loops and further evaluated R-loop dynamics and epigenetic signatures in the retina. Our study underscores the importance of RBPs and R-loop formation as key NRL-interacting regulators of gene expression in retinal rod photoreceptors.

Results

RNA binding proteins constitute a large cohort among NRL interactors

To identify new NRL protein partners that mediate its function in rod photoreceptors, we first conducted an unbiased protein interaction screening. This involved a GST-NRL affinity purification using bovine retina nuclear extracts. This strategy was followed by co-immunoprecipitation (IP) from NRL-enriched high molecular mass fractions of bovine retina, NRL co-IP from mouse retina, and yeast-two-hybrid assays with a NRL domain bait against a human retina “prey” library (schematically shown in **Fig. 1A** [↗](#)).

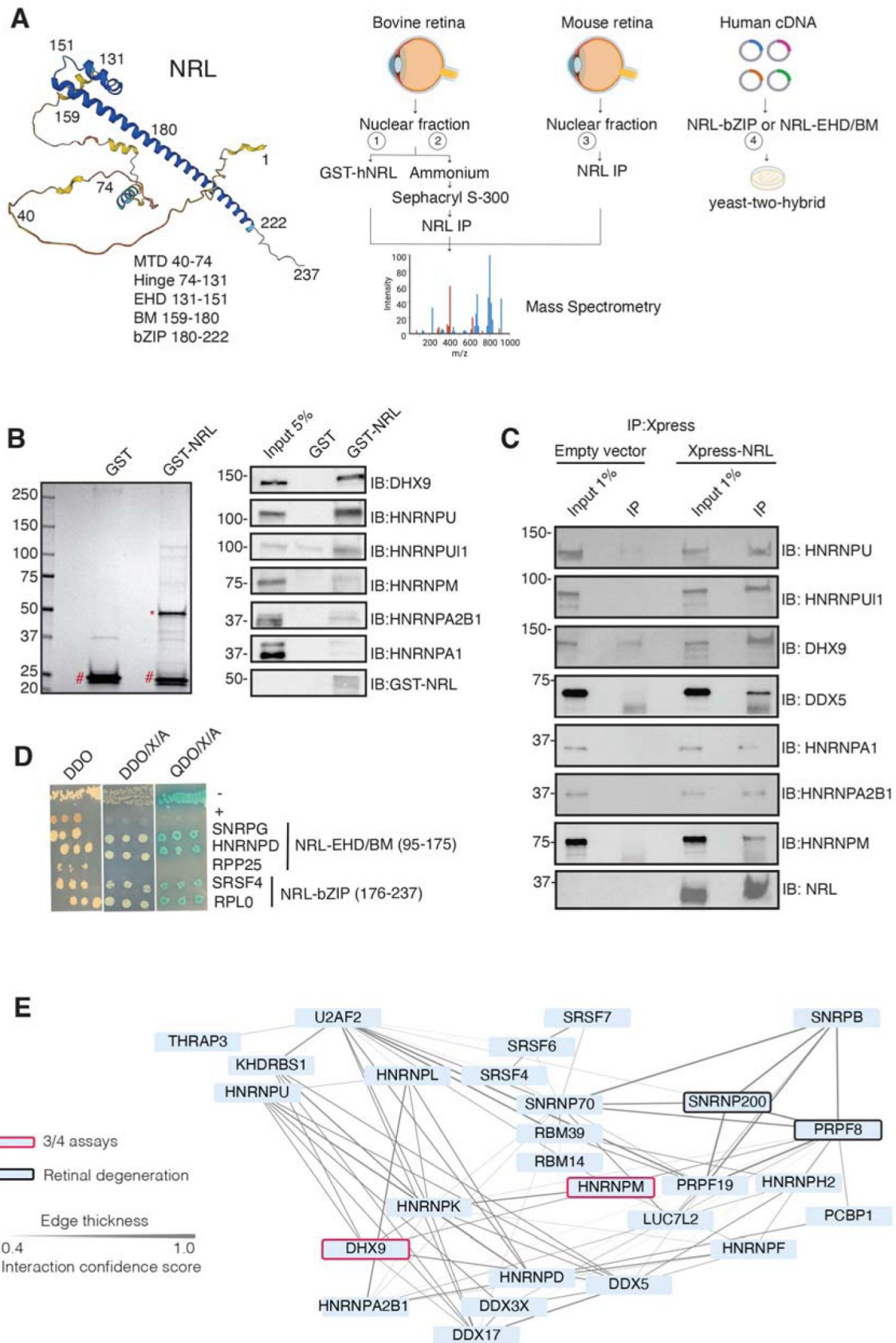


Figure 1.

NRL interacts with RNA-binding proteins.

A) Summary of experimental strategies used to identify NRL interactors. An AlphaFold-predicted model of human NRL is shown, with protein domains highlighted. MTD = minimal transactivation domain, EHD = extended homology domain, BM = basic domain, bZIP = basic leucine zipper. Four different assays were performed to identify NRL interactors. Affinity purifications with Glutathione and NRL antibodies were performed from bovine and mouse retinal lysates and subjected to mass spectrometry. Yeast-two-hybrid experiments from human retina cDNA using NRL bZIP and EHD/BM domains were also performed. Eye and plasmid depictions were obtained from BioRender.com. B) Coomassie staining showing proteins from bovine retina purified with human NRL fused to GST (GST-NRL;*). Purified GST (#) was used as control. Experiments were performed three times with different retinal lysates. Western blot of RBPs identified by LS/MS harboring >10 times enrichment in at least one GST-NRL replicate compared to controls is shown to the right. C) Western blot showing detection of different RBPs co-immunoprecipitating with NRL in HEK293 cells overexpressing Xpress-tagged NRL. Empty vector containing Xpress tag was used as control. D) Yeast colonies from yeast-two-hybrid screens showing positive interaction between RBPs and NRL extended homology domain (EHD) and basic leucine zipper (bZIP) domain. Colonies were plated against controls on SD/- Leu/-Trp (Double Dropout; DDO), SD/-Trp/-Leu/X-alpha-gal/Aureobasidin-A (DDO/X/A) and SD/-Trp/-Leu/-Ade/-His/X-alpha-gal/Aureobasidin-A (QDO/X/A) plates. P53 and Lamin were used as positive and negative controls, respectively. E) PPI network showing RBP experimental interactions from String. Proteins represent a subnetwork of NRL-interacting RBPs found in two out of four assays summarized in A. The edge thickness represents the confidence score with a cutoff of 0.4. Proteins identified in 3 out of 4 assays are highlighted with a red border. Proteins with known causative mutations for inherited retinal degeneration are shown with a black border.

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We identified 28 proteins that were significantly enriched (>2-fold, $p < 0.05$) compared to control samples in three independent GST-NRL pull-down experiments (**Figure 1B**, Table S1). Most proteins were not detected in any of the GST control samples. RBPs represented over 40% of NRL interactors in this set (12/28) (Table S1). We confirmed 6 of the proteins exhibiting the strongest signals by immunoblot analysis of GST-pulldown proteins using specific antibodies (**Fig. 1B**). Notably, interaction of NRL with DHX9, HNRNPU or HNRNPUL1 remained robust even after incubation of the nuclear lysate with Benzonase, a promiscuous nuclease that digests all nucleic acid species, whereas HNRNPA1, HNRNPM and HNRPA2B1 were not or barely pulled down after treatment, indicating a requirement of RNA for the latter set of interactions (Fig. S1A). Removal of RNA significantly increased the background binding for most proteins; however, GST-NRL always maintained stronger signals than controls. This includes HNRNPU and HNRNPUL1 that display white bands indicative of substrate depletion due to high protein levels (Fig. S1A).

Independently and concurrently, immunoprecipitation using anti-NRL antibody of NRL-enriched high molecular mass fractions from the bovine retina, followed by mass-spectrometry analysis, identified several RBP interactors in addition to known interacting proteins such as CRX, NR2E3 and NonO/p54^{nrb} (Table S2). Prioritization of RBPs was done by selecting those with a combined PSM higher than 10 and at least two-fold enrichment. Notably, the most enriched proteins included DHX9, HRRNPU and HNRNPM. Additionally, NRL co-IP experiments using mouse retina nuclear extracts followed by mass spectrometry further validated these findings (Table S2). We also confirmed the interaction between NRL and several of the identified proteins by transfection of tagged constructs in HEK293 cells overexpressing human NRL (**Fig. 1C**). A limited yeast two-

hybrid screening using the NRL bZIP or extended homology domain (EHD) domain baits also identified several RBPs, including SNRPG, HNRNPD, SRSF4, and RPLP0, which were confirmed by positive bait and prey interactions (blue color) (**Fig. 1D** [↗](#)).

Candidate NRL-interacting RBPs, obtained from all four assays ($n = 197$), were annotated with curated protein-protein interactions (PPI) obtained from String (<https://string-db.org/>) [↗](#) ³⁷ to generate a PPI network (Fig. S1B). This network displayed highly interconnected nodes that may indicate interactions with NRL in different cellular contexts. Notably, 30 of these proteins were identified in at least two assays, and two of these, DHX9 and HNRNPM, in three assays (Figure S1B). Of these, twenty-seven RBPs show a high degree of interaction and form a highly interconnected subnetwork (**Figure 1E** [↗](#)). Notably, two of the RBPs (PRPF8 and SNRNP200) [↗](#) ³⁸ have been implicated in photoreceptor degeneration and harbor substantial interactions in this subnetwork (**Figure 1E** [↗](#), black rectangle). We also noted that some RBPs in the network (DHX9, DDX5, DDX17, DDX3X) are RNA helicases with prominent roles in the regulation of R-loops, which are non-B DNA structures comprising of RNA-DNA hybrids and displaced single-stranded DNA with multiple regulatory roles in transcription [↗](#) ³⁹. We then identified 67 high-confidence R-loop binding proteins from four independent studies [↗](#) ^{40–43} by determining shared proteins in their respective R-loop proteomes. Strikingly, 50% of NRL-interacting RBPs were also identified in this high confidence R-loop protein group indicating a high enrichment of R-loop-associated RBPs among NRL interactors (Fisher exact test $p = 2E-13$) (Figure S1C, Table S3).

NRL interacts with the R-loop helicases

DHX9 and DDX5 in rod photoreceptor nuclei

We further focused on the interaction of NRL with R-loop resolvases DHX9 (identified in all affinity purifications) and DDX5 (identified in endogenous NRL-pull downs from bovine and mouse retinas). DHX9 and DDX5 are shown to be present in the same protein complexes [↗](#) ⁴⁴. Notably, immunofluorescence of mouse retina reveals the localization of DHX9 and DDX5 in the rod nuclear periphery, which corresponds to the euchromatin compartment of murine rods (Fig. S2A). We therefore studied DHX9 and DDX5 interaction with NRL *in situ* using a proximity ligation assay (PLA). In mouse retina, DHX9 is observed in all nuclei (**Fig. 2A** [↗](#)). PLAs using anti-NRL and DHX9 antibodies show robust positive interaction signal in the outer nuclear layer (**Fig. 2B** [↗](#)) where NRL is normally expressed (**Fig. 2A** [↗](#)). In contrast, no signal above background is detectable when PLA is performed in *Nrl* KO retina (**Fig. 2B** [↗](#)), suggesting the specificity of PLA signals, even though DHX9 is still expressed throughout the KO retina nuclei (**Fig. 2A** [↗](#)). DDX5 also shows widespread expression throughout the retina (**Fig. 2C** [↗](#)) but displays a lower yet specific interaction signal with NRL in the outer nuclear layer by PLA (**Fig. 2D** [↗](#)). This signal is absent in *Nrl* KO retina (**Fig. 2D** [↗](#)). Similar to mouse retina, DHX9 and DDX5 are detected in all nuclear layers of the human retina whereas NRL is present only in the outer nuclear layer of photoreceptors (**Fig. 2E,G** [↗](#)). Consistent with the results in mouse, NRL-DHX9 PLA signal is clearly detected in the outer nuclear layer of the human retina (**Fig. 2F** [↗](#)), indicating a robust and reproducible interaction between NRL and DHX9. However, NRL-DDX5 PLA signals do not seem to be above negative control (**Fig. 2H, I** [↗](#)). To further ensure that the signal is specific for DHX9, negative controls using human retinas were performed with DHX9 antibodies in combination with Goat IgG. Some background is observed outside the nucleus and in the extracellular space, but no signal enrichment is detected in the photoreceptor nuclear layer (Fig. S2B). Interestingly, in HEK293 cells overexpressing human NRL, PLA signals are robust for both DHX9 and DDX5 throughout the nuclear compartment (**Fig. 2J** [↗](#)). DHX9 and HNRNPU serve as a positive control for PLA and are enriched mostly in the nuclear periphery (**Fig. 2J** [↗](#)). Taken together, these findings suggest a strong interaction between NRL and DHX9 throughout the nuclear compartment in the retina and that a transient and/or more regulated interaction of NRL with DDX5 may require additional protein partners.

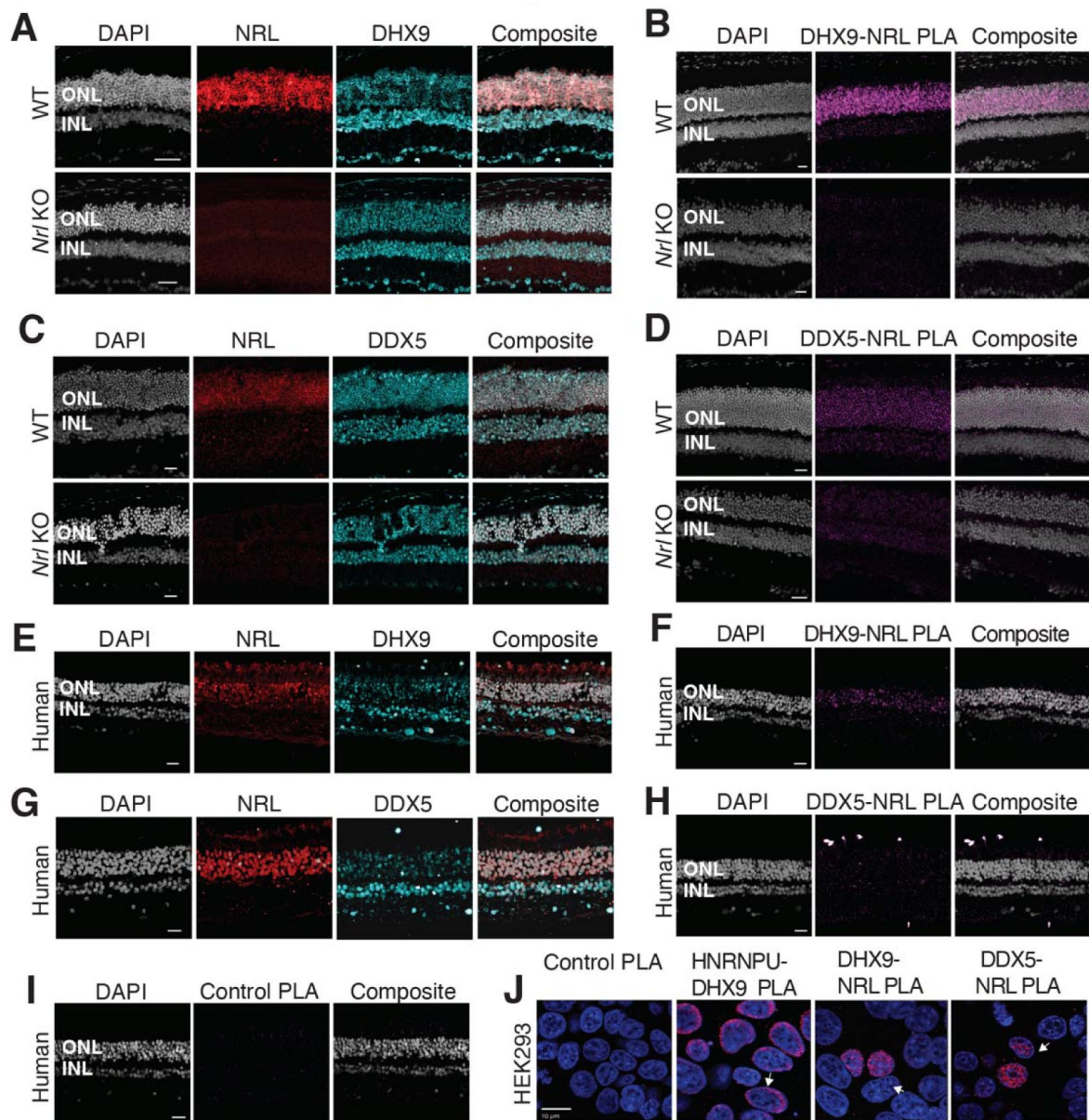


Figure 2.

DDX9 and DDX5 are expressed in retinal photoreceptors and interact with NRL within the nuclear compartment.

A) DHX9 (blue) and NRL (red) expression in adult mouse wild type (WT) and *Nrl* Knockout (KO) retina. B) Proximity ligation assay (PLA) signal (magenta) using anti-DHX9 and NRL antibodies in adult mouse WT and *Nrl* KO retina. C) DDX5 (blue) and NRL (red) expression (blue) in adult mouse WT and *Nrl* KO retina. D) PLA signal (magenta) using anti-DDX5 and NRL antibodies in adult mouse WT and *Nrl* KO retina. E) DHX9 (blue) and NRL (red) expression in adult human retina. F) PLA signal (magenta) using anti-DHX9 and NRL antibodies in the adult human retina. G) DDX5 (blue) and NRL (red) expression in adult human retina. H) PLA signal (magenta) using anti-DDX5 and NRL antibodies in the adult human retina. I) PLA signal (magenta) using no primary antibody in the adult human retina. J) PLA signal in HEK293 cells overexpressing human Xpress-NRL. DHX9 interaction with its known protein partner HNRNPU is shown in the nuclear periphery (arrow). Xpress-NRL interaction with DDX5 and DHX9 in euchromatin is shown in red (arrows). Nuclei were counterstained with DAPI (grey in human and mouse retina; blue in HEK293 cells). Scale bar = 20 μ m. ONL = Outer nuclear layer; INL = Inner nuclear layer.

NRL regulates *DHX9* gene expression

As transcription factors often operate on positive feedback loops, we studied whether the gene expression of NRL interactors was influenced by NRL itself. We examined the regulation of NRL-interacting RBPs by inspecting the published NRL ChIP-Seq and super enhancer profiles from the human retina⁴⁵. Interestingly, over 50% of RBP genes possess NRL ChIP-Seq peaks and almost 25% harbor super enhancers (SE) (Fig. 3A). Fig. S3A shows a few examples of NRL binding to super enhancers regions at *DDX5*, *HNRNPU*, *DDX3X* and *THRAP3* genes. We then focused on the *DHX9* locus and could identify NRL binding peaks in both human and mouse promoter regions (Fig. 3B, Fig. S3B). In concordance, we observe lower levels of *Dhx9* transcripts in *Nrl* KO photoreceptors²⁶ (Fig. 3C). A search of NRL ChIP-Seq peaks²⁵ at the human *DHX9* promoter identified a canonical NRL binding site that is partially conserved in mouse, in concordance with NRL Cut&Run data from mouse retina²⁷ (Fig. 3D). Using a probe containing this sequence, we demonstrate binding of bovine retina nuclear proteins by Electrophoretic Mobility Shift Assays (EMSA) (Fig. 3D). This binding is not detected by mutations in and around the putative NRL-binding site (Fig. 3E). Taken together, our data suggest that NRL can bind to and influence the expression of *Dhx9* and probably other genes encoding RBP interactors.

Interaction between NRL and DHX9 is regulated by R-loops

We further characterized whether the interactions between NRL and DHX9 or DDX5 are regulated by different nucleic acids in HEK293 cells and/or bovine retina. We therefore performed co-IP experiments with antibodies against NRL, DHX9 and DDX5 with or without RNase A, RNase H and DNase. IP using a DDX5 antibody failed to consistently pull down NRL from HEK293 and bovine retina extracts, suggesting that only a small pool of NRL is involved in this interaction, or their interaction is weak and transient. In HEK293 cells overexpressing NRL, the interaction of DHX9 and NRL is enhanced upon treatment of lysates with RNase A (Fig. 4A, C), indicating a negative regulation by RNA. We observed a similar phenomenon in reciprocal pull downs with DHX9 (Fig. 4B, D). On the other hand, treatment with RNase H, which digests RNA only in the DNA:RNA hybrid context, reduced NRL's interaction with both DHX9 and DDX5 suggesting a positive impact of DNA:RNA hybrids (Fig. 4A, C). The reciprocal pull downs with DHX9 did not reveal this change (Fig. 4B, D) probably since NRL accounts for only a small part of DHX9 interactome. Pull down of DHX9 and DDX5 from bovine retina extract with endogenous NRL was not altered by RNase A treatment but was decreased upon removal of DNA:RNA hybrids (Fig. 4E, G), suggesting a role of R-loops in regulating NRL interactions. As in HEK293 cells, reducing DNA:RNA hybrids did not affect the amount of NRL that was pulled down from bovine retina with DHX9 antibody (Fig. 4D, F). Notably, the binding between DHX9 and DDX5 was decreased when RNase A was present (Fig. 4F and S4A), indicating a positive regulatory role of RNA in the retina. Yet, the binding between DHX9 and DDX5 was reduced by RNase H treatment in HEK293 cells, but not in the bovine retina as observed in case of NRL interactions (Fig. 4B, F and S4B). DNA did not influence NRL interactions as evidenced by treatments of lysates with DNase I, which has only 1% activity on DNA:RNA hybrids (Fig. 4A-D).

To assess whether NRL and DHX9 bind to retinal R-loops, we isolated R-loops from enzymatically digested retina gDNA and performed DNA-RNA immunoprecipitation (DRIP) using the S9.6 antibody. After incubations with mouse retinal nuclear lysates, NRL is enriched in R-loops along with DHX9 (Fig. 5A). A pool of NRL also binds to the RNase H-treated sample, possibly reflecting its ability to interact with genomic DNA sequences that may still be present in the pull down. However, DDX5 does not appear to bind to retinal R-loops, suggesting that DDX5 association with R-loops is weaker, transient, or more regulated in the retina, and in agreement with co-IP findings from RNase H treated bovine retina lysates. We then overexpressed GFP-tagged RNase H1 in HEK293 cells to reduce the R-loop levels, as shown previously⁴⁶, and co-transfected the cells

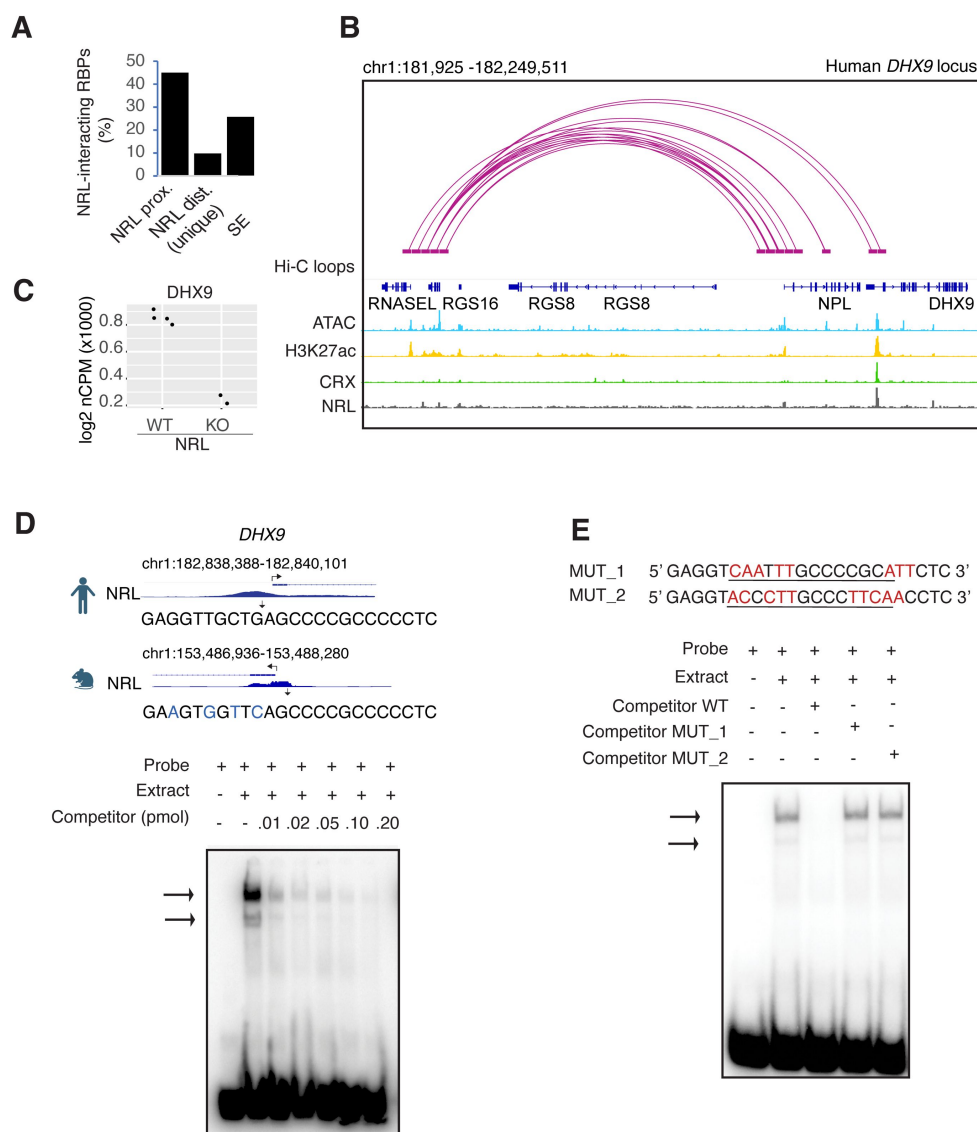


Figure 3.

NRL genetically interacts with RBPs.

A) Bar graph showing the fraction of NRL-interacting RBPs that harbor NRL proximal (< +/-1Kb gene body) or distal (>1 Kb gene body) ChIP-Seq peaks and/or super-enhancers (SE) in the human retina (Data from *Marchal et al. 2022*). B) Genomic view of the human *DHX9* locus showing Hi-C loops, ATAC-Seq, H3K27ac, CRX-ChIP-Seq and NRL-ChIP-Seq tracks (Obtained from *Marchal et al. 2022*). C) Expression levels of *Dhx9* in wild-type and *Nrl* knockout flow-sorted photoreceptors (obtained from *Kim et al., 2016*). D) EMSA autoradiography using a probe containing an NRL motif identified at NRL-ChIP-Seq peak on the human *DHX9* promoter. Chip-Seq or CUT&RUN profiles for human and mouse NRL at the *DHX9* promoter (Obtained from *Marchal et al. 2022* and *Liang et al. 2022*, respectively) are shown. The location of the NRL motif is shown with arrows. The sequence of the ³²P-labeled probe containing human NRL motif (underlined) and its homologous sequence in mouse is shown on the top panel (blue letters indicate nucleotide differences). The blot depicts the formation of specific bands (arrows) after incubation with bovine nuclear retina extracts. Competition assays were performed using unlabeled probes at increasing concentrations (pmol) as shown. E) EMSA autoradiography showing competition assays with 0.2 pmol WT and mutant *DHX9* probes, MUT_1 and MUT_2 (sequences are shown in top panel; nucleotide changes are shown in red).

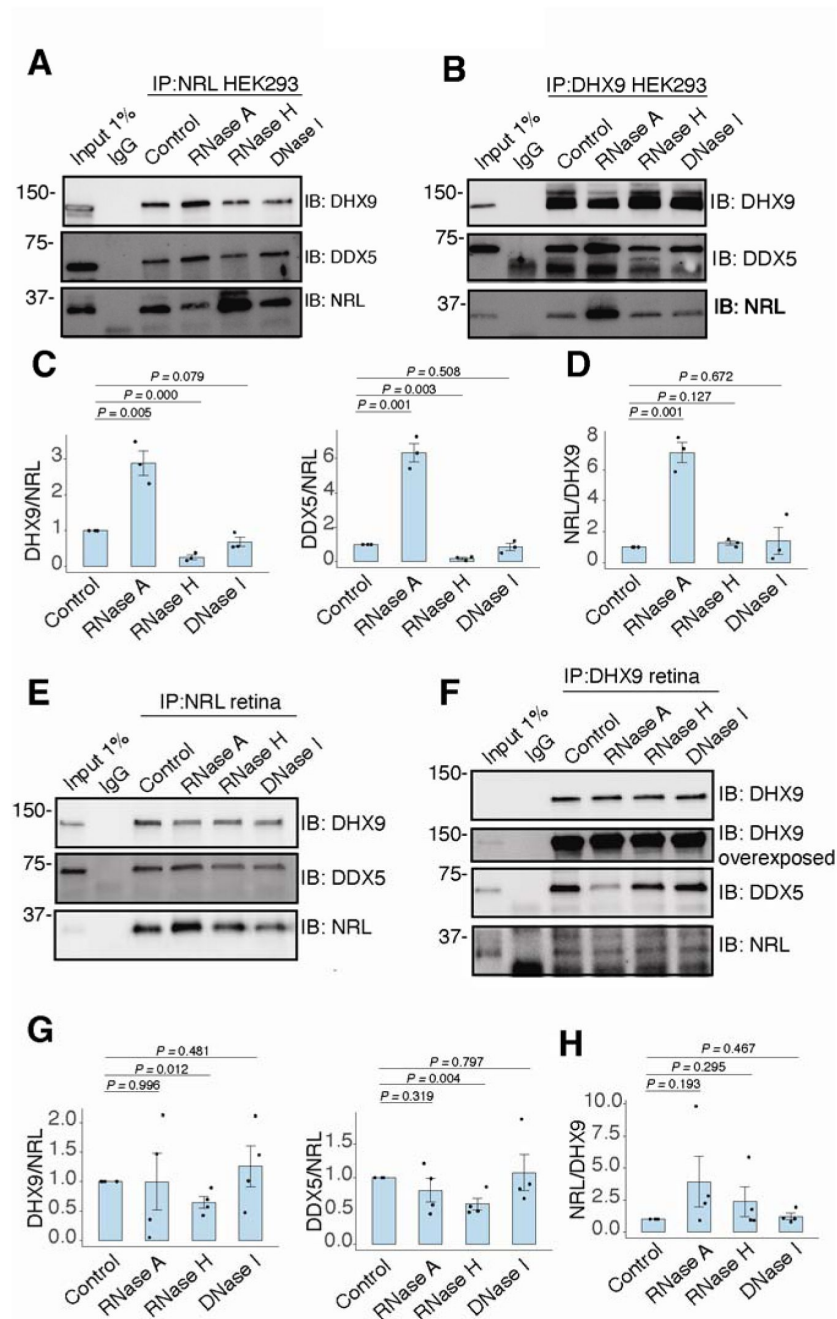


Figure 4.

RNA:DNA hybrids regulate the interaction between NRL and DDX5/DHX9.

A-B) Co-immunoprecipitation (co-IP) of DDX5 and DHX9 from HEK293 cells overexpressing NRL. Lysates were treated for 30 min with different nucleases (as shown) before incubations with respective antibodies. Immunoprecipitation (IP) of NRL (A) or DHX9 (B) and immunoblot (IB) staining for NRL, DHX9 and DDX5 is shown. C-D) Quantification of signal intensities normalized to precipitated NRL and DHX9 (shown in A and B, respectively), ($n = 3$). Data are presented as the mean $\pm$ SEM. An unpaired two-tailed t-test was performed to compare the means of samples against controls. E-F) Co-IP of DDX5 and DHX9 from nuclear fractions of bovine retinas. Lysates were treated for 30 min with different nucleases (as shown) before incubations with respective antibodies. Immunoprecipitation (IP) of NRL (E) or DHX9 (F) and immunoblot (IB) staining for NRL, DHX9 and DDX5 is shown. G-H) Quantification of signal intensities normalized to precipitated NRL and DHX9 (shown in E and F, respectively), ($n = 4$). Data are presented as the mean $\pm$ SEM. Unpaired two-tailed t test was performed to compare means of samples against controls.

with human NRL. We detect a reduction in the number of cells displaying NRL-DHX9 nuclear PLA signals in cells that also harbor wild-type nuclear RNase H1 as compared to the cells having catalytically inactive version of RNase H1 (GFP-dHR) (**Fig. 5B, D**). In addition, we observe a small percentage of cells with PLA signals enriched in subnuclear compartments, especially in cells with wild-type RNase H1 (**Fig. 5B, E**, arrows). Notably, the localization of NRL or DHX9 is not influenced by the expression of RNase H1 (**Fig. 5C**, Fig. S4C). Thus, NRL and DHX9 interaction in the nucleoplasm is favored by R-loops, and R-loops appear to influence the stabilization and localization of NRL-DHX9 complexes.

R-loops are enriched in neuronal genes and display distinct epigenetic signatures

NRL's interplay with R-loops and R-loop proteins suggests a yet unexplored role of R-loops in regulating retinal transcriptional programs. Quantification of R-loop levels using the S9.6 antibody reveals an increase in R-loops with postnatal retinal maturation (**Fig. 6A**). This is in contrast with the relatively stable levels of total expressed genes ($\sim 15,000 > 1$ count per million (CPM)) and transcripts ($\sim 35,000 > 1$ RPKM (Reads Per Kilobase per Million mapped reads)) during retinal development⁴⁷. Total R-loop levels increase in the *Nrl* KO retina, indicating a role of R-loops in controlling cell type-specific gene expression and/or the role of NRL in regulating R-loop levels (Fig. S5).

To identify genomic elements associated with R-loops, we performed single-strand DRIP (ssDRIP) followed by sequencing with four independent adult mouse retina samples and used corresponding RNase H-treated controls. Remarkably, all four samples and all four RNase H-treated samples clustered separately by their PC1 value (Fig. S6A, PCA), demonstrating the specificity of the assay. We identified 4,677 R-loops in our ssDRIP-seq data that were absent in RNase H-treated samples (Fig. S6B and C). We assessed the strand-specificity of these R-loops by quantifying the enrichment of the ssDRIP-seq signal over the opposite strand (**Fig. 6B** and S6B). Our analysis uncovered 3,352 loops without any strand-specificity (hereafter referred as unstranded) and 1,328 strand-specific R-loops (hereafter referred as stranded). Notably, the DRIP-seq coverage was similar for both unstranded and stranded R-loops (Fig. S6C). The identified R-loops intersected with different intergenic and genic regions and are overrepresented in intergenic and promoter regions (**Fig. 6C** and S6D). Intriguingly, stranded R-loops are also enriched at pseudogenes, whereas unstranded R-loops are only enriched at intergenic regions (**Fig. 6C**). In agreement, stranded R-loops are preferentially observed around the transcription start site (TSS), but unstranded R-loops are further downstream of the transcription termination site (TTS) (Fig. S6E).

We then studied whether certain genes/pathways display more R-loops in the retina. We found that stranded R-loops were particularly enriched at neuronal genes associated with synapse function (**Fig. 6D**) and unstranded R-loops were also enriched at genes associated with G protein-coupled receptor signaling (Fig. S6F). In addition, R-loops were identified in 20 genes involved in retinal disease including *Ush2a*, *Pcdh15*, and *Abcc6* (Table S4).

Subsequently, we examined the association of R-loops with different chromatin marks using published retina datasets¹² (**Fig. 6E**). We observed that stranded and unstranded R-loops displayed differences in their epigenetic signatures. Stranded R-loops harbor histone marks associated with active transcription including H3K36me3, H3K4me2, and H3K4me3. However, unstranded R-loops were only enriched with H3K9me3, associated with heterochromatin (**Figure 6E**). Next, we investigated whether NRL binds to R-loops and found that NRL is enriched at both stranded and unstranded R-loops, suggesting a strong association of NRL to R-loops regardless of the epigenetic context (**Fig. 7A**). Notably, other chromatin binding factors such as BRD4, CTCF and Pol II were primarily associated with stranded R-loops. In addition, NRL binding was associated

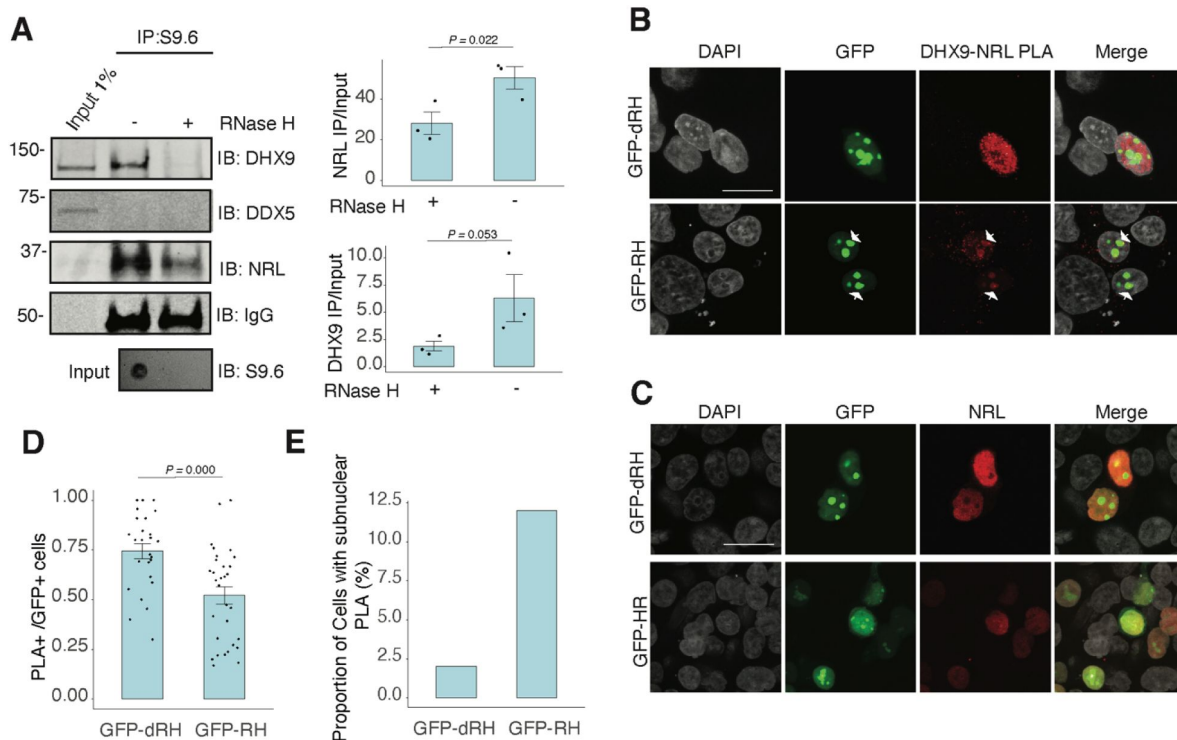


Figure 5.

Nuclear R-loops regulate the interaction between NRL and DHX9.

A) Western blot of DNA:RNA hybrid immunoprecipitation (DRIP) from adult mouse retina showing immunoblot (IB) staining for DHX9, NRL, and DDX5. Retinal genomic DNA (gDNA) was digested with MseI, DdeI, AluI, MboI, incubated with RNase III with/without RNase H and immunoprecipitated with S9.6 antibody/protein G beads. Retinal nuclear lysates were incubated with antibody/bead complexes. Quantification of signal intensities of immunoprecipitated DHX9 and NRL compared to input (n = 3). Data are presented as the mean $\pm$ SEM. Unpaired one-tailed t test was performed to compare means of samples against controls. B) Confocal image of HEK293 cells transfected with NRL and wild type (WT) human RNase H1 or D201N catalytic dead mutant EGFP fusions (GFP-HR, and GFP-dRH, respectively). PLA signals using antibodies for NRL and DHX9 are shown in red. Some cells displayed nucleolar-like accumulation of PLA signal (arrows). C) Confocal image of HEK293 cells transfected with NRL and GFP-dRH or GFP-RH and stained with antibodies against NRL (red). Nuclei are stained with DAPI (grey). Scale bar is 20 μ M. D) Quantification of cells with positive PLA signals from B. Each dot represents a ratio of number of GFP+ cells with nuclear PLA signals per image. Data are presented as the mean $\pm$ SEM. Unpaired two-tailed t test was performed to compare means of samples against controls. E) Bar graph showing percentage of EGFP+ cells harboring NRL-DHX9 PLA signals in subnuclear compartments from B. Cells were counted in four independent assays (n = 83 and 85 cells for GFP-dRH and GFP-RH, respectively).

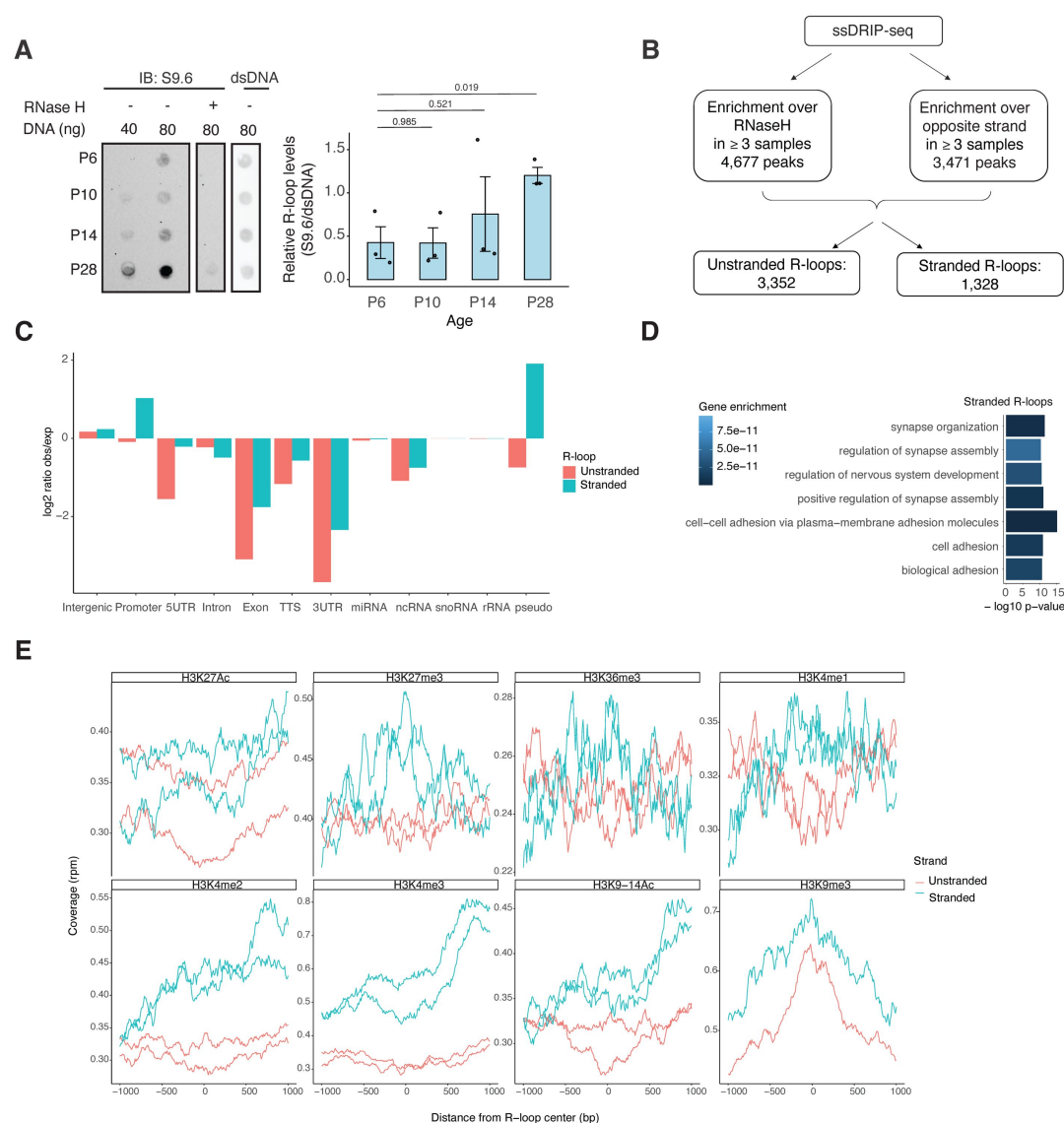


Figure 6.

R-loops are dynamic in the mouse retina and associate with distinct epigenetic signatures.

A) Dot blot of DNA:RNA hybrids from retinal gDNA. Retinas were dissected from mice at different developmental stages as shown. Genomic DNA (gDNA) was treated with RNaseIII with and without RNase H overnight. R-loops were detected using S9.6 antibody (n = 3). Data are presented as the mean $\pm$ SEM. Unpaired two-tailed t test was performed to compare means of samples against controls. B) R-loop peaks from ssDRIP-Seq were identified using RNase H-treated samples as controls. R-loops found in at least 3 samples with a q-value < 0.001 in the narrow call or in the broad call were merged. Stranded R-loops were filtered using the opposite strand as reference. C) Observed vs expected ratio of unstranded and stranded R-loops at different genomic regions. D) Biological process enrichment of genes associated with stranded R-loops. E) Metaplot of H3K27ac, H3K27me3, H3K36me3, H3K4me1, H3K4me2, H3K4me3, H3K9-14Ac, H3K9me3 signals centered on stranded and unstranded R-loop peaks.

with a higher proportion of R-loops at low and high expression genes (**Fig. 7B**, Fig. S7A). We then looked specifically at the NRL-regulated genes²⁷ (Table S5). Overall, NRL target genes have an enrichment of stranded R-loops at the promoter/TSS and unstranded R-loops on the gene body compared to all genes annotated in Ensembl (Figure S7B). Select examples of genes displaying NRL occupancy and harboring stranded or unstranded R-loops are shown in **Fig. 7C**. R-loops can also overlap with proximal promoter or intergenic regions that do not show NRL binding (Fig. S7C). Conversely, R-loops are undetectable in few highly expressed NRL-regulated genes such as *Rhodopsin* (Figure S8). Finally, we studied whether R-loop formation is linked to the gene length. Curiously, the genes without any R-loop are shorter than those overlapping with R-loops, independently of their expression level (**Fig. 7D**). In genes showing low to mid level expression, a concomitant formation of stranded and unstranded R-loops is associated with longer genes compared to the genes with stranded only or unstranded only R-loops (**Fig. 7D**). Taken together, these data demonstrate distinct associations between R-loop types, genomic features, and chromatin binding factors including NRL, which could play a role in the regulation of cell-type specific gene expression.

Discussion

RBP are part of numerous nucleoprotein complexes that participate in diverse cellular processes including transcription of genes in appropriate spatiotemporal context^{48,49}. Several RBPs interact with chromatin and function as integrators of transcription and co-transcriptional RNA processing⁵⁰. We currently have poor understanding of how ubiquitously expressed RBPs are recruited to specific genomic loci at defined locations and in distinct cell types. Here, we demonstrate the interaction of retinal rod photoreceptor-specific bZIP transcription factor NRL with multitude of RBPs that are associated with RNA splicing as well as resolution of R-loops, which form during the transcription process. Focusing specifically on two helicases – DHX9 and DDX5, we show that their interaction with NRL is dependent on specific nucleic acids, and in the case of DHX9 on R-loops. Notably, R-loops are not uniformly present in genes that are highly expressed in the retina but are detected in heterochromatin regions and in lowly-expressed genes. Additionally, R-loops are enriched in genes encoding neuronal synaptic proteins. Our results suggest specific resolution and regulation of R-loops and provide insights into potential mechanisms by which RBPs and R-loops control transcriptional states in the retina.

Most R-loops are formed during transcription and play key roles in transcriptional initiation, elongation and termination. R-loops are reportedly sufficient to initiate transcription of antisense lncRNA from enhancers and other genomic regions⁵¹. R-loops also contribute to Pol II pausing at promoters and in gene bodies, thereby influencing transcriptional outputs⁵². Accumulation of R-loops at polyadenylation-dependent termination regions⁵³ and R-loop-prone sequences downstream from poly A cleavage sites can cause transcriptional termination⁵⁴. Additionally, R-loops may indirectly control gene expression by influencing chromatin structure and the epigenome⁵⁵. In this study, we detect enrichment of R-loops at promoter and intergenic regions indicating their potential role in the modulation of distal and internal regulatory elements. We also demonstrate the presence of R-loops in genes irrespective of their expression levels but also linked to their length, suggesting that R-loops are contributors to the complex regulation of long genes.

We are intrigued by the enrichment of R-loops in neuronal genes associated with synaptic structure and function, including those involved in cell-cell communication. As R-loops are generated during transcription, their relatively stable detection at genes encoding several neuronal and retina-disease associated proteins indicates distinct transcriptional control mechanisms for different gene families. One plausible explanation is slower turn-over of RNA and requirement for smaller protein amounts. The greater likelihood of neuronal genes to harbor R-loops may also be because of their larger gene size⁵⁶. R-loops may also contribute to regulation

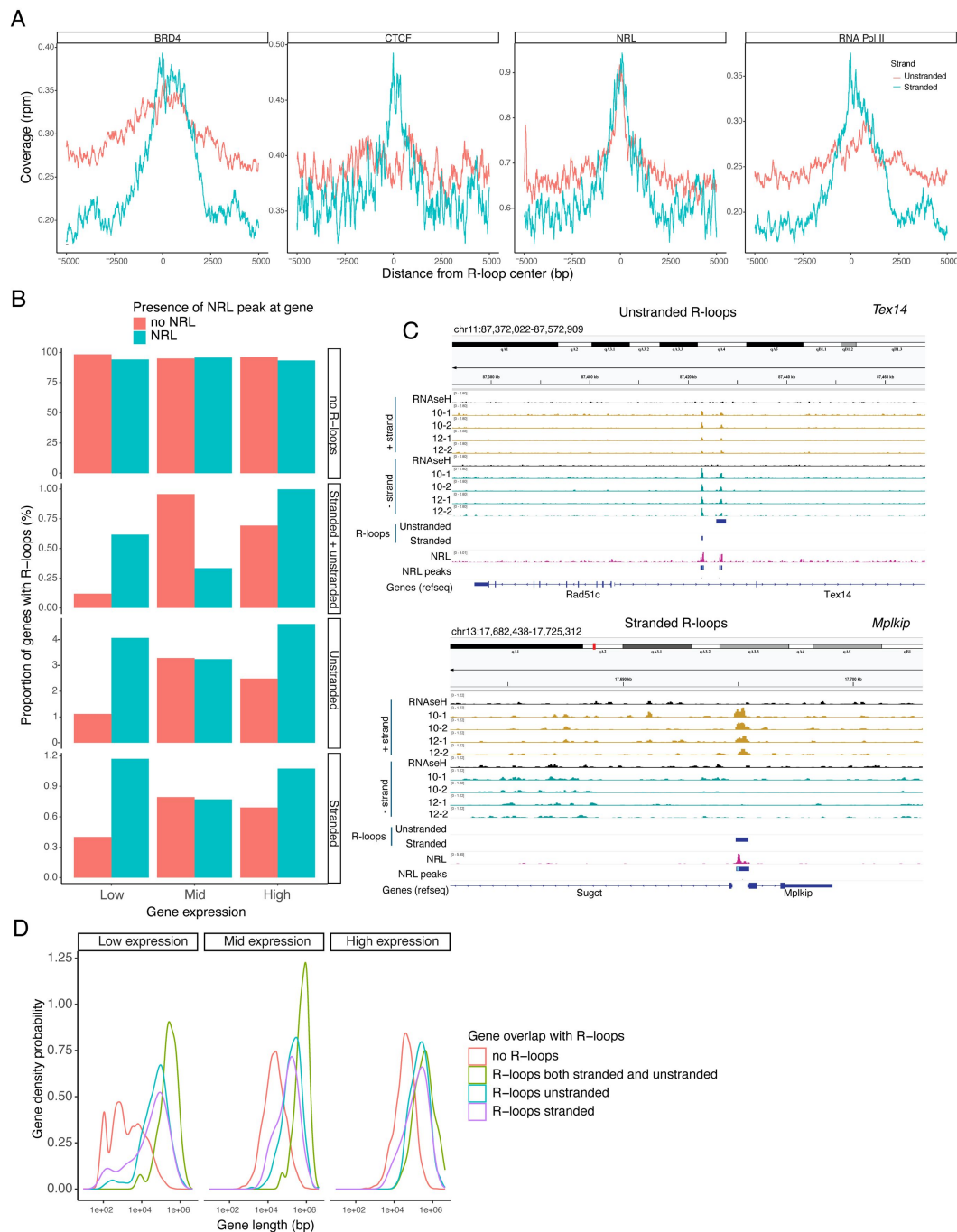


Figure 7.

NRL is associated with different types of R-loops.

A) Metaplot of BRD4 (SRR4252658), CTCF (SRR4252685), NRL (Cut&Run), and RNA pol II (SRR4252922) signals centered on stranded and unstranded R-loop peaks. B) Proportion of genes with and without stranded and unstranded R-loops and harboring NRL Cut&Run and ChIP-Seq peaks. Genes were divided into low, mid and high according to their expression level. C) Genome view of *Tex14* and *Mplkip* mouse genes displaying ssDRIP-Seq signal in four retinas. Signals are shown for the positive (orange) and negative (blue) strands separately. RNase H treated samples are pooled and shown for each strand. Peak calls for NRL and unstranded and stranded R-loops are shown in blue. D) Gene density probability plot showing the distribution of R-loops over expressed genes of different lengths. Genes were divided into three groups according to their expression levels.

of long neural genes as is the case for DNA damage⁵⁷. Further investigations into R-loop formation and resolution at distinct genomic regions can offer mechanistic insights into gene regulation as well as non-coding variations relevant to disease.

Our results indicate that unstranded R-loops (containing RNA:DNA hybrids on both strands) are enriched in intergenic regions and occur downstream of genes, overlapping the repressive mark H3K9me3. This observation suggests that bidirectional transcription is associated with the formation of repressive genomic regions. Indeed, R-loops are also reported to be involved in maintenance of heterochromatin and telomers^{58,59}. The potential role of R-loops in maintaining a repressed chromatin state in the retina warrants further investigation. We note that R-loops near transcriptional termination sites have also been associated with the formation of repressive epigenetic marks and termination of transcription. For example, R-loops can induce antisense transcription at gene termination sequences, leading to the generation of double-stranded RNA and recruitment of G9a histone methyltransferase⁶⁰. Bidirectional transcription termination by Pol II also offers a potential mechanism of termination at convergent genes⁶¹. R-loops localizing near H3K9me3 sites are shown to be associated with regulation of long genes⁶². Intriguingly, NRL genomic occupancy overlaps with unstranded R-loops at the gene body, suggesting a functional link that may be unrelated to transcriptional initiation. Additional experiments are needed to understand the relationship between R-loops and repressive marks in postmitotic retinal neurons.

NRL regulates the transcriptional state of many rod-specific genes²⁶. The absolute requirement of NRL for rod photoreceptor cell fate determination and its continued high expression for functional maintenance of rods strongly indicate a multitude of roles of NRL beyond transcription initiation. Interactions of NRL with RBPs and R-loops are likely significant contributors to gene regulation. In particular, we note that NRL-interacting RNA helicases, DHX9 and DDX5, have a direct role in controlling R-loop resolution and cell fate specification⁴¹, DNA damage response⁶³ and splicing regulation⁶⁴. The interaction of NRL with DHX9 is very robust in GST pull downs, in HEK293 cells, and in bovine retina, suggesting that NRL-DHX9 complexes might play key roles in rod photoreceptors. Curiously, the interaction of NRL with DDX5 is weaker, suggesting that it may be more regulated or context dependent. In agreement, DHX9 but not DDX5 is enriched in retinal R-loops. Removal of RNA in bovine retina did not weaken the interaction between NRL and both DHX9 and DDX5, indicating that RNA is not required for many NRL-RBP interactions. Thus, interaction of NRL with R-loop resolvases and with R-loops shown here *in vitro* through R-loop pull downs, and *in vivo* through DRIP-Seq, suggest that NRL contributes to R-loop regulation in retinal photoreceptors. Indeed, NRL-regulated genes are enriched for stranded R-loops at promoter regions and unstranded R-loops at gene bodies implying distinct modes of R-loop-mediated gene regulation. Notably, dynamics of R-loops is proposed to regulate somatic cell reprogramming that involves interaction of SOX2 with DDX5⁴¹. **Fig. 8** shows a proposed model of NRL-mediated regulation of R-loops during gene expression in rod photoreceptors. Our studies thus provide an avenue to investigate the role of R-loops in retinal development and functional maintenance.

Excess of R-loops can generate DNA damage, genomic instability, and directly activate immune response^{46,65}. Dysregulation of R-loops has been implicated in a variety of diseases including neurodegeneration⁶⁶. Even during normal aging, aberrant transcription contributes to mis-splicing and increased R-loop formation in *Drosophila* photoreceptors⁶⁷. As R-loops are low or absent at many key retina genes, failure of the mechanism that maintains these genes free of R-loops could contribute to retinal disease. Interestingly, monoallelic rare *DHX9* variants increase R-loop levels and are shown to be associated with neurological disorders⁶⁸. Therefore, further investigation into NRL-DHX9 interaction with R-loops in the retinal photoreceptors could uncover mechanisms that maintain transcriptional homeostasis and genomic stability.

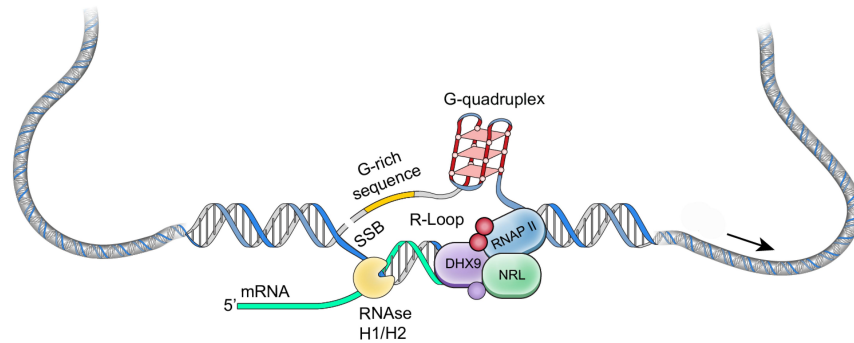


Figure 8.

A putative model of R-loop regulation in retinal rod photoreceptors.

We hypothesize that NRL guides R-loop resolvases such as DHX9 at photoreceptor genes to critically regulate gene expression. SSB, single strand break.

Transcription factors are reported to be involved in controlling alternate splicing ⁷⁰[70](#). Based on our discovery of NRL's interaction with splicing factors and R-loop binding proteins, we hypothesize a potential role of NRL in determining patterns and rates of splicing, which is prevalent in photoreceptors ⁶⁹[69](#)–⁷³[73](#). NRL-RBP interactions may also influence the rate of transcription elongation and/or termination and consequently transcript isoform diversity in rod photoreceptors. Notably, mutations in splicing factors as well as aberrant splicing of many genes have been associated with retinal degeneration ⁷⁰[70](#),⁷⁴[74](#),⁷⁵[75](#).

Finally, our results indicate that other RNA-dependent chromatin processes may be influenced by NRL-RBP complexes. Many of the identified RBPs participate in maintaining chromatin architecture and modifications and in DNA damage repair, and their functions require further exploration. Overall, we propose that RBPs play key roles in rod photoreceptors through their interactions with NRL. Future studies will aim at unravelling the physiological significance of NRL-interacting RBPs on R-loops and chromatin regulatory mechanisms in photoreceptors.

Methods

Mouse strains and husbandry

All procedures involving mice were approved by the Animal Care and Use Committee of the National Eye Institute (NEI-ASP#650). C57BL/6J (B6) mice were kept in a 12 hr light/12 hr dark cycle and fed ad libitum at the NEI animal facility. Both male and female mice were used in this study.

Human tissue

Human donor eyes were procured from Lions World Vision Institute (Tampa, FL). Autopsy material from unidentified deceased individuals is not subject to Institutional Review Board (IRB) review and does not need a determination from the Office of Human Subjects Research Protections (OHSRP) according to 45 CFR 46 and NIH policy (OHSRP ID#: 18-NEI-00619). Eyes were enucleated within 6 hr of death, followed by making a 2 cm incision at the limbus before immersing in a solution of 4% formaldehyde in PBS. Eyes were subsequently transferred to PBS for shipment and kept at 4°C. Before sectioning, eyes were fixed in 4% formaldehyde in PBS for 2 hr and immersed in 30% sucrose.

Antibodies

The following antibodies were used: DDX5 Monoclonal antibody (Cat. No. 67025-1-Ig), DHX9 Polyclonal antibody (Cat. No. 17721-1-AP), and HNRNPA2B1 (Cat. No. 14813-1-AP), all obtained from Proteintech (Rosemont, IL, USA); HNRNPA1 (Cat.No. 8443, Cell Signaling, Danvers, MA, USA); DNA-RNA Hybrid antibody S9.6 (Cat. No. 65683, Active Motif, Carlsbad, CA, USA); Xpress Monoclonal Antibody (Cat. No. R910-25, ThermoFisher, Waltham, MA, USA); HNRNPM (Cat. No. A6937, Abclonal, Woburn, MA, USA); HNRNPU and HNRNPU11 (Cat. No. MA1-24632 and Cat. No. 10578-1-AP, respectively, ThermoFisher, Waltham, MA); rabbit anti-NRL (Swaroop lab ²⁷[27](#) and R&D systems Cat. AF2945); double stranded DNA (Cat. No. MAB030, MilliporeSigma, Burlington, MA, USA).

Plasmids

Human NRL constructs in pcDNATM4/HisMax vector containing an N-terminal Xpress tag (Cat. No. V86420, ThermoFisher Scientific, Waltham, MA, USA) or in pGEX-4 T2 plasmid containing an N-terminal GST tag (GE Healthcare/Cytiva; cytivalifesciences.com ⁷⁶[76](#)) have been described previously ⁷⁶[76](#). Human RNase H1 or D201N catalytic dead mutant EGFP fusions (GFP-HR, and GFP-dHR) were developed in the Cimprich lab ⁴⁶[46](#) and obtained from Addgene (Cat. No.196702, and 196703).

Mass spectrometry

Affinity-purified protein samples after incubation of GST-NRL (n = 3 for GST-NRL; n = 3 for GST only control) with bovine (n = 2 for NRL immunoprecipitation, IP; n = 1 for IgG control) or mouse retina (n = 2 for NRL IP, n = 1 for IgG control) were subjected to Liquid Chromatography by Tandem Mass Spectrometry (LC-MS-MS) analysis (Poochon Scientific, Frederick, MD, USA). Samples were digested with trypsin, peptides were extracted, desalted and analyzed using Q-Exactive hybrid Quadrupole-Orbitrap Mass Spectrometer and Thermo Dionex UltiMate 3000 RSLCnano System (ThermoFisher Scientific, Waltham, MA, USA). Peptides were ionized and sprayed into the mass spectrometer using Nanospray Flex Ion Source ES071 (Thermo Scientific, Waltham, MA, USA). Raw data files were searched against bovine, mouse or human protein sequence databases (National Center for Biomedical Information, NCBI) using Proteome Discoverer 1.4 software (Thermo Scientific, Waltham, MA, USA) based on SEQUEST algorithm. Carbamidomethylation of cysteines was set as a fixed modification, whereas oxidation and deamidation Q/N-deamidated (+0.98402 Da) were set as dynamic modifications. The minimum peptide length was specified to be five amino acids. The precursor mass tolerance was set to 15 ppm and fragment mass tolerance to 0.05 Da. The maximum false peptide discovery rate was specified as 0.01. All assembled proteins with peptides sequences and matched spectrum counts (peptide spectrum match counts (#PSM)) were included in the analysis.

Yeast two-hybrid (Y2H) screening

We performed Y2H assays using a modified Matchmaker System (Clontech, Mountain View, CA, USA) with a human retinal cDNA prey library pGADT7 vector (containing Gal4 activating Domain). Two partial NRL domains were synthesized by Genewiz (Germantown, MD, USA): (1) NRL Extended Homology Domain (EHD) with the Basic Motif (BM) (residues 95-175), and (2) partial NRL BM with NRL bZIP domain (residues 176-237). NRL domain containing inserts were subcloned into the bait vector, pGBKT7 (containing Gal4 binding domain), and transformed into Y2HGold yeast. The yeast containing a bait were inoculated at 30°C overnight in 50 mL (-)Trp broth to obtain a cell yield of 7.5×10^8 cells per culture. Subsequently, yeast cells were pelleted at $3000 \times g$ for 5 min, resuspended in 50 mL of (-)Trp broth and incubated at 30°C for 3-4 hr with shaking (225 rpm). Yeast colonies containing baits were transformed with 10 µg of human retinal prey cDNA library. For screening, positive colonies were selected on SD/-Trp/-Leu/X-alpha-gal/Aureobasidin-A (DDO/X/A) media to determine β-galactosidase activity and antibiotic resistance as an indicator of a positive interaction with NRL. These interactors were validated by patch plating onto higher stringency SD/-Trp/-Leu/-Ade/-His/X-alpha-gal/Aureobasidin-A (QDO/X/A) media. Positive interactor inserts were PCR amplified, sequenced using T7 primer, and identified by Blastn and Blastx (NCBI).

To confirm NRL interacting RBPs, colonies containing RBP prey and NRL domain baits were resuspended in ultra-pure water, and plated against controls on 5 different plating media: SD/-Trp, SD/-Leu, SD/-Trp/-Leu, SD/-Trp/-Leu/X-alpha-gal/Aureobasidin-A and SD/-Trp/-Leu/-Ade/-His/X-alpha-gal/Aureobasidin-A.

Cell culture and transfection

HEK293 cells were cultured in Dulbecco's modified Eagles's medium (Cat. No. 11885084, ThermoFisher Scientific, Grand Island, NY, USA) Containing 10% fetal calf serum (Cat. No. S11550, R&D Systems, Flowery Branch, GA, USA), 100-units/mL penicillinG and 100 µg/mL streptomycin (Cat. No. 15140122, ThermoFisher Scientific, Grand Island, NY, USA). Cell transfection was performed with X-tremeGENE 9 DNA Transfection Reagent (Cat. No. 6365779001, Cat. No. 06365787001, Roche, Mannheim, Germany) or Lipofectamine 2000 (Cat. No. 11668027, ThermoFisher Scientific, V Carlsbad, CA, USA) per manufacturer's instructions.

Immunofluorescence and microscopy

Eyes from *Nrl* WT or *Nrl*-knockout (KO) C57BL/6 mice ¹⁸ at postnatal day (P) 28 were fixed in 4% paraformaldehyde for 1 hr, washed 3x in PBS, and embedded in optimal cutting temperature (OCT) medium on dry ice. For NRL detection and proximity ligation assay (PLA), eyes were directly embedded in OCT. Human eyes were fixed for 2 hr in 4% PFA, washed as above and embedded in OCT. Eyes were cryosectioned at 14 μ m and mounted on SuperFrost Plus slides (Thermo Fisher Scientific, Waltham, MA, USA). For NRL detection and PLA in mouse retinas, fresh sections were postfixed in 4% PFA for 7 min. Retinal sections were then incubated in blocking solution (5% bovine serum albumin, 0.3% Triton X-100 in PBS) for 1 hr followed by primary antibody (anti-DDX5, anti-DHX9 or anti-NRL 1:100) incubation in blocking solution at 4°C overnight. After three washes of 10 min each in PBS, sections were incubated in secondary antibody in blocking solution in the presence of the DNA dye DAPI (4',6-diamidino-2-phenylindole) for 1 hr. Sections were washed 3x in PBS and mounted in ProLong® Gold Antifade Reagent (Life Technologies Inc., Carlsbad, CA, USA). Confocal images were acquired using SP8 Leica 2-photon confocal microscope or a Zeiss LSM 800 point scanning confocal microscope using the AiryScan detector.

Proximity ligation assay (PLA)

PLA was carried out using Duolink® PLA Fluorescence kit (Millipore, Sigma) per manufacturer instructions. Briefly, HEK293 cells were transfected with NRL-Xpress or RNase H1 constructs (GFP-HR and GFP-dHR). Cells were washed with PBS 48 hr post-transfection, fixed using 4% paraformaldehyde in PBS for 15 min at room temperature and washed three times with PBS. Mouse and human retinas were fixed as described above. Cells or sections were permeabilized using 0.3% Triton X in PBS for 5 min. Duolink blocking solution was added for 60 min at 37°C. Rabbit anti-DHX9, anti-DDX5 antibodies, anti-NRL or Xpress antibodies were diluted in Duolink antibody diluent and incubated overnight at 4°C. After washing, plus and minus PLA probes were diluted 1:5 into Duolink antibody diluent. Cells were incubated at 37°C for 1 hr. After further washing, the ligation was performed for 30 min at 37°C.

Protein lysates and enzymatic treatments

Bovine Sephacryl S-300 fractions

High molecular mass complexes containing NRL were obtained by size exclusion chromatography as described earlier ⁷⁶. Briefly, bovine retinal nuclear extracts were fractionated using 40% ammonium sulfate. Isolation of high molecular mass protein complexes was performed using HiPrep 16/60 Sephacryl S-300 High-Resolution column (GE Lifesciences/Cytiva). The peaks corresponding to 650 and 450 kDa containing NRL were immunoprecipitated using NRL antibody/protein A bead complexes overnight at 4°C, as described earlier.

Bovine nuclear extracts for GST purification

Retinas were resuspended in CE1 buffer (10 mM HEPES-KOH pH 7.5, 1.5 mM MgCl₂, 10 mM KCl, 10% Glycerol, 1 mM DTT) with protease inhibitors (PI) for 10 min at 4°C. Retinas were homogenized in dounce homogenizer 25 times and pelleted at 250×g for 10 min. Nuclear pellets were resuspended in high salt buffer (450 mM NaCl, 10 mM Tris pH 7.4, 2 mM EGTA, 0.1% Triton X-100, 2 mM MgCl₂) with PI for 30 min at 4°C. Chromatin was pelleted at 10,000×g for 10 min. Supernatants were diluted to 150 mM NaCl and incubated with BSA-blocked antibody-bead conjugates overnight as described earlier. For RNase A treatments, lysates were incubated with RNase A at 1 μ g per μ g DNA for 30 min at 37°C before antibody incubation.

Bovine and HEK293 nuclear extracts and treatments

HEK293 cells cultured in 6-well plates as described earlier, were lysed in 1 mL co-immunoprecipitation (co-IP) binding buffer (40 mM Tris-HCl pH 8.0, 150 mM NaCl, 2 mM EDTA and 0.2% NP-40) containing protease inhibitor cocktail. The nuclear pellet from bovine retina was obtained as described above and resuspended in 1 ml of co-IP binding buffer. After sonication using an ultrasonic liquid processor (MISONIX, Inc, Farmingdale, NY) until complete chromatin dissolution, samples were clarified by centrifugation at 16,000×g for 10 min at 4°C. The DNA concentration in the lysate was measured by Qubit dsDNA BR Assay Kit (Thermo Fisher Scientific, Waltham, MA). For each immunoprecipitation, 100 µl lysate was treated as follows: For control, the lysate was mixed with 50 units of RNaseOUT recombinant ribonuclease inhibitor (Thermo Fisher Scientific, Waltham, MA, USA); for RNase A treatment, the lysate was mixed with 5 µg of RNase A per µg of DNA (Thermo Fisher Scientific, Waltham, MA, USA); for RNase H treatment, 2.5 units of RNase H were added per µg of DNA with 1x RNase H buffer (New England Biolabs, Ipswich, MA); for DNase I treatment, 1 unit of DNase I was added per µg of DNA with 1x DNase I reaction buffer (Thermo Fisher Scientific, Waltham, MA). All reactions were performed for 30 min at 37°C. After treatments, the lysates were incubated with antibodies as described above.

Co-immunoprecipitation (co-IP)

Dynabeads Protein A (Thermo Fisher Scientific, Waltham, MA) were blocked with 1% BSA at room temperature for 1 h and incubated with 1 µg of the respective antibodies for another hour. After washing off the unbound antibody, the lysates were added to the beads and incubated overnight at 4°C. The beads were then washed 3 times for 15 min each with co-immunoprecipitation wash buffer (150 mM NaCl, 10 mM Tris-HCl pH 7.4, 2 mM EGTA, 1% Triton X-100, 2 mM MgCl₂) and bound protein complexes were eluted by boiling in Laemmli SDS sample buffer for 10 min at 95°C.

GST affinity purification

GST-NRL was expressed in BL21(DE3) competent cells (ThermoFisher Scientific, Waltham, MA) using 0.2 mM IPTG for 16 hr at 20°C and purified using Glutathione Sepharose beads (Cat. No. GE17-0756-01, MilliporeSigma, Burlington, MA) by incubating BSA-blocked beads with cell lysates for 3 hr at 4°C. GST-NRL beads were then incubated for 2 hr at 4°C with nuclear fraction from bovine retina (prepared as above) with or without 25 U of benzonase. Afterwards, the beads were washed three times for 10 min each in PBS with 1% Triton X-100, then boiled in Laemmli sample buffer containing 100 mM DTT for 7 min and stored at -80°C or loaded into 4-15% acrylamide gels for immunoblotting.

R-loop detection

R-loops were extracted as previously reported [77](#) with some modifications. Retinas or HEK293 cells were incubated in nuclear extraction buffer (Tris pH 7.4 10 mM, NaCl 10 mM, MgCl₂ 3 mM, BSA 1%, Tween-20 0.1 %, NP-40 0.1%) for 10 min. After dounce homogenization, nuclei were pelleted at 500×g for 5 min at 4°C. Genomic DNA was obtained by incubating nuclei in 10 mM Tris-HCl containing 1 mM EDTA (TE) with 1% SDS and 60 µg Proteinase K20 overnight at 37°C. DNA was extracted with Phenol/Chloroform Isoamyl alcohol 25:24:1 (PCI). Briefly, equal volume of PCI was added to each sample. After gentle mixing, samples were spun down at 12,000×g for 30 min at 4°C. DNA was precipitated by adding 1/10 volume of 3 M NaOAc, pH 5.2 and 2.5 volumes of 100% (vol/vol) ethanol. The pellet was washed with 1 mL of 70% ethanol. After gentle mixing, samples were centrifuged at 12,000×g for 15 min at 4°C. The supernatant was discarded, and the pellet was air dried and resuspended in TE buffer. DNA was incubated with RNase H buffer and RNase III at 37°C OV with and without RNase H.

To detect R-loops during development, retinas were collected from three B6 mice at each time point (postnatal day (P)6, P10, P14 and P28).

Dot blot analysis

gDNA samples diluted in TE buffer were blotted on a positively charged Hybond® -N+ hybridization membrane (Thermo Fisher Scientific, Waltham, MA, USA) using a dot-blot apparatus (Cat. No. 1706545, Bio-Rad Laboratories, Hercules, CA, USA). Membranes were crosslinked using 1200 µJ UV light.

Electrophoretic Mobility Shift Assay (EMSA)

DNA oligonucleotides (5 pmoles, forward strand) were incubated with 50 µCi of ³²P-ATP and 10 units of T4 polynucleotide kinase for 30 min at 37°C. The reaction was terminated by heating at 65°C for 10 min and column purified to remove unincorporated label. This labeled oligonucleotide was annealed to 5 pmoles of the complementary strand by heating to 95°C for 5 min and cooling slowly to room temperature. Double stranded ³²P-labeled oligonucleotides (0.2 pmol, 20,000 cpm) were incubated with 5 µg of bovine nuclear and 1X Binding Buffer (LightShift, Thermo Fisher Scientific), 50 ng/µl Poly dI-dC, and 5 mM MgCl₂ for 1 hr. Reactions were terminated by addition of loading buffer, separated on 8% DNA retardation gel (Invitrogen, Waltham, MA, USA). Subsequently, the gel was dried and exposed to Phosphor screen overnight, which was scanned using Typhoon FLA 9500 Biomolecular Imager (GE Lifesciences). Cold competition assays were performed by preincubating unlabeled double stranded oligonucleotides at 0.1 to 0.2 pmole concentration, as indicated, with the nuclear extract for 5 min before the addition of the labeled probes. Mutant probes were incubated at 0.2 pmole.

Probes for EMSA were designed by examining putative NRL binding motifs in ATAC-Seq footprints overlapping with NRL ChIP-Seq peaks as described ⁴⁵.

- Probe for DHX9: 5' GAGGTTGCTGAGCCCCGCCCTC 3'.
- Mutant DHX9 probe 1: 5' GAGGTCAATTGCCCCGATTCTC 3'.
- Mutant DHX9 probe 2: 5' GAGGTACCCTTGCCCTTCAACCTC 3'.

Protein Network

RBPs included in the analysis had >2-fold increase compared to the controls and were present at 10 or more total peptide spectrum matches (PSMs) in experimental samples. Protein-protein interactions (PPI) among RBPs were extracted from STRING Database (<https://string-db.org/>). Only data identified experimentally was included for analysis. The confidence threshold was set to 0.4 and displayed in the network as edge thickness. Networks were created using Cytoscape (<https://cytoscape.org/>).

Gene regulation analysis

Normalized transcripts per million (nTPM) of neuronal cells from single-cell human data (<https://www.proteinatlas.org/>) was used to compare expression levels in photoreceptors and other neuronal cells. Overlaps between genes encoding RBPs (+/-1 kb from gene body) identified in two out of four assays with human retina NRL-ChIP-Seq peaks and super-enhancers from human retina (obtained from ⁴⁵) were identified using bedtools (<https://bedtools.readthedocs.io/en/latest/>). Genomic views of gene loci were obtained using IGV (<https://software.broadinstitute.org/software/igv/>).

S9.6 DNA-RNA co-immunoprecipitation

DRIP was performed as described ^{41,78}, with minor modifications. Genomic DNA was digested with 50 U of Mse I, Dde I, Alu I, and Mbo I. Digested DNA (500 ng) was treated with RNase III and/or RNase H overnight as above. gDNA was added to Protein G Dynabeads (Thermo scientific)

preincubated with 1.5 µg of S9.6 antibody for 1 hr at RT in 500 µl of binding buffer (10 mM Na₂HPO₄, 140 mM NaCl, 0.05% Triton X-100). After three washes in binding buffer, R-loop-antibody-protein G complex was incubated with mouse retina lysate (200 µg) overnight at 4°C. Nuclear lysates were obtained as described above and pretreated with RNase A (0.1 ng per 1 µg DNA) for 1 hr at 37°C followed by incubation with 1000 U RNaseOut for 10 min at 4°C. After 3 washes with IP buffer, S9.6-bound complexes were analyzed by immunoblotting.

Single-stranded DNA-RNA immunoprecipitation (ssDRIP)-Seq

ssDRIP-seq was performed as described earlier⁴¹. The libraries were prepared using the Accel-NGS 1S Plus DNA Library Kit (IDT technologies Coralville, IA, USA). Briefly, the DRIPed DNA sample obtained as above was fragmented to an average length of 250 base pairs (bp) utilizing an S220 Focused-ultrasonicator (Covaris, Woburn, MA, USA). The fragmented DNA was then denatured by heating at 98°C for 2 min, followed by immediate cooling on ice for 2 min. The R2 adapter was first ligated to the 3' end of the ssDNA, followed by an extension step. Subsequently, the R1 adapter was ligated to the 5' end. After PCR amplification, the resulting libraries were purified using AMPure XP beads (Beckman Coulter, San Jose, CA). The quality of each library was assessed with a TapeStation instrument (Agilent, Palo Alto, CA, USA), and sequencing was carried out on an Illumina NextSeq 2000 system (San Diego, CA, USA) at 2 × 100 bp at a final library concentration of 1100pM with a sequencing depth of greater than 20 million pair end reads.

After trimming adaptor sequences, low-complexity tails were removed by further trimming 10 bps from each read on the 3' end of the first read in the pair and 5' end of the second read in the pair. Reads were aligned to the mouse genome (mm10) using Bowtie2. Duplicated reads, reads with score < 20 (including multi-mapping reads) and reads mapping ENCODE consortium blacklisted regions of mm10 genome (github.com/Boyle-Lab/Blacklist/) were removed. Mapped reads were split by strand using samtools. To confirm the quality of the experiment, a PCA was run on the 500bp coverage scaled in R for all samples treated or not with RNase-H, using R ggfortify package. R-loops calling was performed using MACS2. First, all R-loops (without regard of any strand specificity) were called on all reads, using the merged RNase H treated samples as control dataset using MACS2 with the default parameters (narrow peaks) and the broad peaks option. R-loops found in at least 3 samples with a q-value ≤ 0.001 in the narrow call or in the broad call were merged. From this list of R-loops, strand specificity was assessed for each R-loop. To do so, peaks presenting signal enrichment from one strand over the other were called on reads split by strand, using the reads mapping the opposite strand as control dataset, similarly to the R-loops call. By intersecting the peaks presenting signal enrichment from one strand over the other and the total R-loops, we extracted the strand-specific R-loops and the unstranded R-loops. The annotation of R loops, gene ontology analysis and chromatin marks enrichment were computed using HOMER and plotted using R package ggplot2. Overlap between gene expression levels, R-loops and NRL peaks were computed using Bedtools. For this, we utilized our previously published expression data that was reanalyzed for GRCm38/ENSv98 annotation⁴⁷, re-analyzed NRL chromatin binding data (ChIP-seq from ref.²⁵ and Cut&Run from ref.²⁷). Overlap between R-loops and NRL target genes²⁷ (Table S5) was computed using Bedtools. The gene promoter-TSS region was defined as -2.5 kb to +0.5 kb from the TSS. The gene downstream region was defined as -0.5 kb to 2.5 kb from the gene end. We generated related plots using the R package ggplot2 and chromatin profiles using IGV.

Statistical analysis

Means between groups were compared using Student's t-test in at least three biological replicates. The differences were considered statistically significant with a two or one-tailed p-value of < 0.05. Immunoblots of pull-downs were analyzed using ImageJ (imagej.net/ij/index.html) by calculating the ratio of signal intensities of prey and bait normalized to controls. Dot blot signal intensities were normalized to input gDNA. R-loop protein enrichment was calculated using the Fisher exact

test with a contingency table based on the total number of retina-expressed nuclear RBPs. Mouse nuclear RBPs data were obtained from the RBP2GO database (<https://rbp2go.dkfz.de/>), which reports 1764 RBPs including 1534 that are expressed in the mouse retina.

Data Availability

All sequencing data that support the findings of this study have been deposited in the National Center for Biotechnology Information Gene Expression Omnibus (GEO) and are accessible through the GEO accession number GSE274666.

All other relevant data are available from the Lead Contact on request.

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Additional information

Author Contributions

Conceptualization, X.C.D. and A.S.; Methodology and Investigation, X.C.D., X.L., K.P.Jr., B.T., M.E., J.N., S.P.Y.; Data analysis, X.C.D., X.L., K.P.Jr., J.N., S.P.Y.; Bioinformatic analysis, C.M.; Writing - Original Draft, X.C.D., A.S.; Writing-Review & Editing, all authors; Funding, X.C.D., A.S.; Supervision and Project administration, A.S.

Additional files

Supplemental Data [↗](#)

Supplemental Table S2 [↗](#)

Supplemental Table S3 [↗](#)

Supplemental Table S4 [↗](#)

Supplemental Table S5 [↗](#)

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Author information

Ximena Corso-Díaz

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States, Department of Ophthalmology, Byers Eye Institute, Stanford University, Stanford, United States

For correspondence: ximenac@stanford.edu

Xulong Liang

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Kiam Preston Jr.

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Bilguun Tegshee

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Milton A English

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Jacob Nellisery

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Sharda Prasad Yadav

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

Claire Marchal

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States, In silichrom Ltd, Newbury, United Kingdom

Anand Swaroop

Neurobiology, Neurodegeneration and Repair Laboratory, National Eye Institute, National Institutes of Health, Bethesda, United States

ORCID iD: [0000-0002-1975-1141](https://orcid.org/0000-0002-1975-1141)

For correspondence: swaroopa@nei.nih.gov

Editors

Reviewing Editor

Xiaorong Liu

University of Virginia, Charlottesville, United States of America

Senior Editor

Lois Smith

Boston Children's Hospital, Boston, United States of America

Reviewer #1 (Public review):

Summary:

In this manuscript, Corso-Diaz et al, focus on the NRL transcription factor (TF), which is critical for retinal rod photoreceptor development and function. The authors profile NRL's protein interactome, revealing several RNA-binding proteins (RBPs) among its components. Notably, many of these RBPs are associated with R-loop biology, including DHX9 helicase, which is the primary focus of this study. R-loops are three-stranded nucleic acid structures that frequently form during transcription. The authors demonstrate that R-loop levels increase during photoreceptor maturation and establish an interaction between NRL TF and DHX9 helicase. The association between NRL and RBPs like DHX9 suggests a cooperative regulation of gene expression in a cell-type-specific manner, an intriguing discovery relevant to photoreceptor health. Since DHX9 is a key regulator of R-loop homeostasis, the study proposes a potential mechanism where a cell-type-specific TF controls the expression of certain genes by modulating R-loop homeostasis. The authors also identify another R-loop resolvase, DDX5 as having weaker interaction with NRL, perhaps due to indirect mechanism.

This is a very interesting study providing genome-wide mapping of R-loops in mammalian retina, which shows an enrichment of R-loops over intergenic regions as well as genes encoding neuronal function factors. The R-loop-enriched genes are longer than genes without R-loops, which supports previous findings from studies in neuronal cells. This is a very relevant study highlighting the possible mechanism of gene expression regulation via interactions between TFs, RNA binding proteins, and R-loops. In that regard, it would be very interesting to uncover the biological relevance of such gene regulation. The authors provide adequate evidence of interaction between R-loops and NRL TF via in vitro IP assay and genomic co-localization, however, this interaction can be mediated via multiple R-loop or RNA-binding proteins. Thus, follow-up studies would be appropriate to characterize this interaction in more detail.

<https://doi.org/10.7554/eLife.103259.2.sa2>

Reviewer #2 (Public review):

Summary:

The Authors utilize biochemical approaches to determine and validate NRL protein-protein interactions to further understand the mechanisms by which the NRL transcription factor controls rod photoreceptor gene regulatory networks. Observations that NRL displays numerous protein-protein interactions with RNA-binding proteins, many of which are involved in R-loop biology, led the authors to investigate the role of RNA and R-loops in mediating protein-protein interactions and profile the co-localization of R-loops with NRL genomic occupancy.

Strengths:

Overall, the manuscript is well-written, providing succinct explanations of the observed results and potential implications. Additionally, the Authors use multiple orthogonal techniques and tissue samples to reproduce and validate that NRL interacts with DHX9 and DDX5. Experiments also utilize specific assays to understand the influence of RNA and R-loops on protein-protein interactions. The Authors also use state-of-the-art techniques to

profile R-loop localization within the retina and integrate multiple previously established datasets to correlate R-loop presence with transcription factor binding and chromatin marks an attempt to understand the significance of R-loops in the retina.

Weaknesses:

In general, the Authors provide interpretations of the data that fit a narrative about NRL and the perceived significance of interactions with RNA binding proteins. Large-scale screens for NRL protein interactions were conducted but all of the data is not reported. For example, NRL IP-Mass Spec was performed, but the authors only provide interaction/detection data for identified interactions with known RNA binding proteins. We cannot assess the enrichment of interactions or specificity of interactions with RNA binding proteins based on the reported results. Additionally, the lack of experiments testing the functional significance of Nrl interactions with R-loops within the developing retina fails to provide novel biological insights into the regulation of gene regulatory networks. While this provides additional avenues for research in the future, it is unclear that NRL interaction with R-loops have physiological relevance for photoreceptor health or function.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this manuscript, Corso-Diaz et al, focus on the NRL transcription factor (TF), which is critical for retinal rod photoreceptor development and function. The authors profile NRL's protein interactome, revealing several RNA-binding proteins (RBPs) among its components. Notably, many of these RBPs are associated with R-loop biology, including DHX9 helicase, which is the primary focus of this study. R-loops are three-stranded nucleic acid structures that frequently form during transcription. The authors demonstrate that R-loop levels increase during photoreceptor maturation and establish an interaction between NRL TF and DHX9 helicase. The association between NRL and RBPs like DHX9 suggests a cooperative regulation of gene expression in a cell-type-specific manner, an intriguing discovery relevant to photoreceptor health. Since DHX9 is a key regulator of R-loop homeostasis, the study proposes a potential mechanism where a cell-type-specific TF controls the expression of certain genes by modulating R-loop homeostasis. This study also presents the first data on R-loop mapping in mammalian retinas and shows the enrichment of R-loops over intergenic regions as well as genes encoding neuronal function factors. While the research topic is very important, there is some concern regarding the data presented: there are substantial data supporting the interaction between NRL and DHX9, including pull-down experiments and proximity labeling assay (PLA), however, the data showing an interaction between NRL and DDX5, another R-loop-associated helicase, are inadequate. Importantly, the data supporting the claim that NRL interacts with R-loops are absolutely insufficient and at best, correlative. The next concerns are regarding the R-loop mapping data analysis and visualization.

Strengths:

There is compelling evidence that the NRL transcription factor interacts with several RNA binding proteins, and specifically, sufficient data supporting the interaction of NRL with

DHX9 helicase.

A major strength is the use of the single-stranded R-loop mapping method in the mouse retina.

Weaknesses:

(1) Figure S1A: There is a strong band in GST-IP (control IP) for either HNRNPUI1 or HNRNPU, although the authors state in their results that there is a strong interaction of these two RBPs with NRL.

Under our experimental conditions, most RNA-binding proteins displayed higher binding to glutathione beads (Fig. S1A). However, GST-NRL purifications showed much stronger signals for respective RBPs. In the case of HNRNPU and HNRNPUI1, white bands that are indicative of substrate depletion due to higher protein levels are observed in GST-NRL lanes. Additionally, in Figures 1B and 1C, there is a clear enrichment of HNRNPU and HNRNPUI1 above the background signal. We added this to the text. See page 5.

Both DHX9 and DDX5 samples have a faint band in the GST-IP.

RNA-binding proteins may display some background as observed in other studies (e.g. PMID: 32704541). We think that showing the raw data without decreasing the exposure time is useful and that there is a clear enrichment compared to controls. In addition, we tested the interaction in multiple systems.

There is an extremely faint band for HNRNPA2B1 in the GST-NRL IP lane. Given this is a pull-down with added benzonase treatment to remove all nucleic acids, these data suggest, that previously observed NRL interactions with these particular RBPs are mediated via nucleic acids. Similarly, there is a loss of band signal for HNRNM in this assay, although it was identified as an NRL-interacting protein in three assays, which again suggests that nucleic acids mediate the interaction.

Thank you for highlighting this point. We mention in the manuscript that the interaction between HNRNPM and A1 depends on nucleic acids, as noted by the reviewer, since there is no obvious band after the pull-down. We have now added that the interaction of NRL with HNRNPA1B1 is likely dependent on nucleic acids as well, given its weak signal. See page 5.

(2) The data supporting NRL-DDX5 interaction in rod photoreceptor nuclei is very weak. In Figure 2D, the PLA signal for DDX5-NRL is very weak in the adult mouse retina and is absent in the human retina, as shown in Figure 2H.

We agree with the reviewer. We think that the signal for DDX5 is weak, and we addressed this in the text. We noted on page 7: “Taken together, these findings suggest a strong interaction between NRL and DHX9 throughout the nuclear compartment in the retina and that a transient and/or more regulated interaction of NRL with DDX5 may require additional protein partners.” We have modified this sentence to add that the data also suggest transient interaction or the requirement of additional protein partners for stable interaction. See page 7.

Given that there is no NRL-KO available for the human PLA assay, the control experiments using single-protein antibodies should be included in the assay. Similarly, the single-protein antibody control PLA experiments should be included in the experimental data presented in Figure 2J.

Thank you for the suggestion. We performed PLAs using both DHX9 and IgG in the human retina and observed no specific amplification signal. Some background is observed outside the nucleus and in the extracellular space. We added these results to the text and to the supplementary information. See page 7 and Fig.S2B.

(3) The EMSA experiment using a probe containing NRL binding motif within the DHX9 promoter should include incubation with retina nuclear extracts depleted for NRL as a control.

In EMSA experiments, we used bovine retina to obtain enough protein quantities. As suggested by the reviewer, using NRL depleted extract would increase the specificity of observed gel shift and complement our pre-immune serum as a negative control. However, removal of all the NRL protein using the antibodies available was not feasible. In the future, we will use enough mice to obtain large quantities of protein for this experiment and will collect retinas from Nrl knockout as negative control.

(4) There is a reduced amount of DHX9 pulled down in NRL-IP in HEK293 cells, but there is no statistically significant difference in the reciprocal IP (DHX9-IP and blotting for NRL) (Figure 4C).

We believe the reviewer is referring to the data in Figure 4C showing that RNase H treatment led to significantly reduced pulldown of DHX9 as compared to control, but the reciprocal IP in Figure 4D showed no statistical significance between control and RNase H treatment. In Figure 4D, we hypothesize that NRL may account for only a small proportion of DHX9's interactome, so the change in NRL levels could not be detected due to the sensitivity of our assay. DHX9 likely constitutes a large proportion of NRL's interactome in HEK293 cells, hence the change in DHX9 level was more obvious when pulling down with NRL. We added this information to the results. See page 8.

(5) The only data supporting the claim that NRL interacts with R-loops are presented in Figure 5A.

Additional evidence that NRL interacts with R-loops comes from DRIP-Seq experiments where signals from R-loops overlap with NRL ChIP-Seq signals (Figure 7A). This shows that R-loops and NRL co-occur on multiple genomic regions. In addition, indirect evidence of NRL and R-loops' interaction is shown in pull down experiments and PLA assays where R-loops influence DHX9 and NRL binding. We clarified this in the discussion. See page 14.

This is a co-IP of R-loops and then blotting for NRL, DHX9, and DDX5. Here, there is no signal for DDX5, quantification of DHX9 signal shows no statistically significant difference between RNase H treated and untreated samples, while NRL shows a signal in RNase H treated sample. These data are not sufficient to make the statement regarding the interaction of NRL with R-loops.

Thank you for this comment. We respectfully disagree as we observe statistically significant enrichment for both NRL and DHX9 in these experiments (See Fig5A). Some NRL continues to bind to DNA that is pulled down nonspecifically, which may be expected since NRL is a transcription factor. See for example R-loop binding by the transcription factor Sox2 (PMID: 32704541). However, binding to R-loops is evidenced by an enrichment compared to RNase H-treated sample. We clarified this in Results section (See page 9).

(6) Regarding R-loop mapping, the data analysis is quite confusing. The authors perform two different types of analyses: either overall narrow and broad peak analysis or strand-

specific analysis. Given that the authors used ssDRIP-seq, which is a method designed to map R-loops strand specifically, it is confusing to perform different types of analyses.

Thank you for highlighting this point. This has enhanced the clarity of the methods and enriched the discussion. We aimed to identify R-loops as accurately as possible. We conducted two types of analyses to capture different aspects of R-loops: one that looks at overall patterns (narrow and broad peaks) and another that focuses on specific strands of DNA.

Using ssDRIP-seq, which is designed to map R-loops on specific strands, allowed us to examine R-loops formed in only one strand and those formed on both strands. To identify strand-specific R-loops, we filtered our RNase-H enriched peaks for those enriched on one strand compared to the opposite strand. We clarified the analysis in the results section, and Figure 6B. See page 10 and methods section page 25.

Next, the peak analysis is usually performed based on the RNase H treated R-loop mapping; what does it mean then to have a pool of "Not R-loops", see Figure 6B?

The "Not R-loop" group refers to peaks called using the opposite strand that are not observed when calling peaks using RNase H as control. We modified this figure for clarity (Figure 6B).

In that regard, what does the term "unstranded" R-loops mean? Based on the authors' definition, these are R-loops that do not fall within the group of strand-specific R-loops. The authors should explain the reasons behind these types of analyses and explain, what the biological relevance of these different types of R-loops is.

Thank you for helping us clarify this point. Unstranded R-loops are DNA regions containing DNA:RNA hybrids on both plus and minus strands and possibly representing bidirectional transcription by Pol II. We observed that unstranded R-loops are enriched only in intergenic regions, H3K9me3 regions, and downstream of the transcriptional termination site (TTS). We added to the discussion the possible implications of these enrichments, including regulation of Pol II termination and transcription of long genes. See Page 13.

(7) It would be more useful to show the percent distribution of R-loops over the different genomic regions, instead of showing p-value enrichment, see Figure 6C.

Since most of the genome is non-coding, plotting the distribution as a proportion was not informative since the vast proportion of the data falls in intergenic regions. However, we created a new figure showing observed vs. expected ratio that seems to be more informative and moved the current p-value figure to the supplement in revised version. See Figure 6C and S6D.

(8) Based on the model presented, NRL regulates R-loop biology via interaction with RBPs, such as DHX9, a known R-loop resolution helicase. Given that the gene targets of NRL TF are known, it would be useful to then analyze the R-loop mapping data across this gene set.

Thank you for this suggestion. We performed an analysis of R-loops on NRL-regulated genes. Interestingly, NRL target genes have an enrichment of stranded R-loops at the promoter/TSS and unstranded R-loops on the gene body compared to all Ensembl genes (Figure S7B). We added a table containing all NRL-regulated genes we used for this analysis (table S5) and a figure showing this result (Fig. S7B).

Reviewer #2 (Public review):

Summary:

The authors utilize biochemical approaches to determine and validate NRL protein-protein interactions to further understand the mechanisms by which the NRL transcription factor controls rod photoreceptor gene regulatory networks. Observations that NRL displays numerous protein-protein interactions with RNA-binding proteins, many of which are involved in R-loop biology, led the authors to investigate the role of RNA and R-loops in mediating protein-protein interactions and profile the co-localization of R-loops with NRL genomic occupancy.

Strengths:

Overall, the manuscript is very well written, providing succinct explanations of the observed results and potential implications. Additionally, the authors use multiple orthogonal techniques and tissue samples to reproduce and validate that NRL interacts with DHX9 and DDX5. Experiments also utilize specific assays to understand the influence of RNA and R-loops on protein-protein interactions. The authors also use state-of-the-art techniques to profile R-loop localization within the retina and integrate multiple previously established datasets to correlate R-loop presence with transcription factor binding and chromatin marks in an attempt to understand the significance of R-loops in the retina.

Weaknesses:

In general, the authors provide superficial interpretations of the data that fit a narrative but fail to provide alternative explanations or address caveats of the results. Specifically, many bands are present in interaction studies either in control lanes (GST controls) of Westerns or large amounts of background in PLA experiments.

We have added additional information to the text regarding the presence of background signals in pull downs. We wish to note that experimental samples always exceeded background signals. We believe that reporting these raw findings (rather than showing shorter exposures) is valuable for the scientific community. We did not observe any background in the proximity ligation assay (PLA) that exceeded what is typically expected, and the signals were clearly discernible. Cases where signals are weaker, such as with DDX5, have been highlighted. In addition, we added a DHX9-IgG negative control for the human PLA experiment. See page 5 and Fig. S2B.

Additionally, the lack of experiments testing the functional significance of Nrl interactions or R-loops within the developing retina fails to provide novel biological insights into the regulation of gene regulatory networks other than, 'This could be a potentially important new mechanism'.

We agree that functional experiments are necessary to understand the molecular mechanisms behind R-loop regulation in the retina; however, we believe it goes beyond the scope of this initial characterization (as this is the first report on R-loops in the retina). We are currently pursuing these studies.

We performed new analysis on NRL-regulated genes as suggested by reviewer 1. We show that NRL target genes have an enrichment of stranded R-loops at the promoter/TSS and unstranded R-loops on the gene body compared to all Ensembl genes (Figure S7B), providing further evidence of the functional interaction between NRL and R-loops. See table S5 and Fig. S7B, and discussion.

Additionally, the authors test the necessity of RNA for NRL/DHX9 interactions but don't show RNA binding of NRL or DHX9 or the sufficiency of RNA to interfere/mediate protein-protein interactions. Recent work has highlighted the prevalence of RNA binding by

transcription factors through Arginine Rich Motifs that are located near the DNA binding domains of transcription factors.

We agree that the role of RNA in these complexes is very exciting, and we are currently pursuing these studies. However, we believe that they fall outside the scope of this initial report on R-loops in the retina.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

There are a couple of minor comments:

(1) Unfinished sentence; page 11, the end of the first paragraph.

Thank you for catching this error. We removed the unfinished text.

(2) Page 6: Figure S2A should be Figure S2.

In general, the manuscript would benefit from a deeper explanation of the biological relevance of R-loop formation and the connection to NRL TF and the expression of genes regulated by NRL. In this regard, a more substantial description of the model would be useful.

We have modified the discussion for clarity and included new ideas on possible roles of R-loops in gene regulation of photoreceptors.

Reviewer #2 (Recommendations for the authors):

(1) The specificity of interactions needs to be addressed:

- Figure 1B - HNRNPU1 bands present in GST control.*
- Figure 1C - Bands present in the Empty Vector control IP for HNRNPU and DHX9.*
- Supplemental Figure 1A - most proteins are present in GST control suggesting prevalent binding to GST and lack of specificity for other interactions.*

Thank you for your comment. RNA-binding proteins can have more background as observed in other studies (e.g. PMID: 32704541) but there is always a higher signal in experimental samples compared to controls. While we agree that we can enhance the conditions for immunoprecipitation (IP) by optimizing washing buffers, exposure and other parameters, we believe the current methods tell the story. We have added additional text explaining this. See page 5.

(2) Use of the term 'Strongest' interaction - IPs don't directly address the strength of interaction, but depend on levels of expression AND affinity. The strength of interaction should be tested using techniques like an OCTET or SPR assay. One can also quantify the effect that RNA would have in such an assay.

Thank you for your suggestion. We replaced the term 'stronger' with “higher signal” and “robust” at most places. The source of protein lysates is the same for experiments and controls, thus the amount of protein is consistent in both conditions, and not dependent on level of gene expression.

(3) In supplemental tables, please use the proper gene names, not the UniProt peptide name. For example, there are no genes named ELAV1-ELAV4. These should be ELAVL1-

ELAVL4. A short glance identifies >10 gene name errors.

Thank you for the suggestion. We updated current gene names in all tables.

(4) Please provide the rationale for the choice of DNA sequence for the DHX9 nucleotide sequence used for EMSA assays. In the human DHX9 locus, the NRL ChIP-seq peak looks to be contained in Intron1 whereas the NRL ChIP-seq peak in mouse DHX9 looks to be in the proximal upstream promoter. Did the authors choose an evolutionarily conserved sequence in the promoter region that contained the NRL motif or does the probe sequence arise from the sequence that has known NRL binding as assayed by NRL ChIP-seq? A zoomed-in image of the NRL ChIP-seq pile-ups in the DHX9 locus in each species would be beneficial.

Thank you for this suggestion. The probe was chosen by scanning for NRL binding motifs on the Chip-Seq peak at the human DHX9 promoter. We added a Zoom-in image of the ChIP-Seq or CUT&RUN reads for NRL on both human and mouse retinas. Figure 3D shows NRL binding in both species in regions containing the homologous motif. The sequence is partially conserved and shown in the figure.

(5) Normalization in RNaseH/RNaseA Co-IP experiments. Why does RNaseH treatment result in increased NRL IP (increased NRL expression?) or does RNaseA treatment cause reduced IP of DHX9? These differences seem to cause a 'denominator' effect, leading the Authors to conclude decreased co-IP of DHX9 with NRL when R-loops are inhibited or increased co-IP of NRL with DHX9 when RNA is degraded. An alternate interpretation would be that inhibiting the R-loop binding of NRL unmasks the epitope for antibody recognition. The authors should test NRL binding to RNA and determine if RNA binding affects the co-IP of NRL with DHX9.

We agree that removing total RNA by RNase A or R-loops by RNase H may alter the accessibility of our antibodies to the epitopes, resulting in the differences in the level of total protein pulled down. However, we quantified the relative level of the associating protein to the total protein and confirmed, in reciprocal assays, that RNase A treatment led to increased interaction between NRL and DHX9. However, the quantification was not consistent between the reciprocal IPs upon RNase H treatment. We reason that in Figure 4D, as NRL may account for only a small proportion of DHX9's interactome, the change in NRL level could not be detected due to the sensitivity of our assay. However reciprocally, DHX9 can constitute a larger proportion of NRL's interactome in HEK293 cells, hence the change in DHX9 level was more obvious. We added this information to the text. See page 8.

(6) Figure 7 - Malat1 - there doesn't seem to be an overlap of NRL with Stranded R-loop peaks in this image. Nrl seems to flank the region of R-loops.

We changed Malat1 for Mplkip that shows a direct overlap of Nrl binding and R-loops. See Figure 7C.

(7) Results end with 'A Model'. Seems like some concluding remarks and references to Figure 8 were mistakenly left out.

Thank you for catching this typo. We removed the misplaced text.

(8) Model and Discussion - authors should show raw data for RHO with respect to NRL binding and R-loops. No evidence was provided regarding R-loops (or lack thereof) in the Rhodopsin locus. Additionally, conclusions stating that "R-loops... are specifically depleted from genes, such as Rhodopsin, with high expression levels" go against Figures

7B and 7C. *Malat1* is one of the highest expressed genes in the retina and contains R-loops.

Thank you for helping us clarify our hypothesis. We added a genome browser view of Rhodopsin showing the absence of R-loops (Fig. S8). We hypothesize that R-loops could interfere with achieving higher rates of transcription, however we did not mean to say that all high expressed genes lack R-loops. We have rephrased the discussion to clarify this point.

(9) Neuronal genes, particularly those involved in synaptic transmission are known to be, on average, longer than most genes (Gabel, 2015; PMID: 25762136). Is it possible that R-loops are detected at genes involved in synaptic function/structure solely because of transcript length, as it takes longer for transcription termination to resolve in genes that are longer? A plot showing R-loop enrichment and transcript length would address this.

We added a plot showing gene length in relation to R-loops and expression levels. We observed that R-loops are more common over long genes regardless of their expression levels. We also observed that the concomitant presence of stranded and unstranded R-loops is restricted to the longest genes in most cases. We added this to Figure 7D.

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